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The explicit hypergeometric-modularity method II

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Abstract

In the first paper of this sequence, Allen et al (Adv. Math 478:110411, 2025), we provided an explicit hypergeometric modularity method by combining different techniques from the classical, p-adic, and finite field settings. In this article, we explore an application of this method from a motivic viewpoint through some known hypergeometric well-poised formulae of Whipple and McCarthy. Using well-poised hypergeometric formulae we construct a class of degree four Galois representations of corresponding cyclotomic fields. These representations are then shown to be extendable to the full absolute Galois group $G_{\mathbb{Q}}$ and the L-function of each extension coincides with the L-function of an automorphic form.

Keywords: Hypergeometric functions, Modular forms, Supercongruence, Galois representations, Character sums, L-values, Well-poised formula

Mathematics Subject Classification: 33C20, 11F03, 11F66, 11F80, 11T24

1 Introduction

Properties of L-series arising from automorphic forms and Galois representations have been one of the central interests in number theory. L-functions arising from geometric objects are expected to be automorphic, implying they can be analytically continued to the whole complex plane. This automorphy has been established in special cases—most famously in the Modularity Theorem [8,37,42]—but remains open in general. In this series of papers, we consider L-series and Galois representations arising from a hypergeometric background. In [30], Roberts and Rodriguez Villegas give a general description of hypergeometric motives which are governed by hypergeometric data and their associated combinatorics, which have been further used in papers including [12,25] and others. As illustrated below in Figure 1, these hypergeometric motives give a uniform perspective on three mutually enriching aspects: the classical complex aspect [3] largely based on the Gamma function $\Gamma(\cdot)$, the finite field or Galois aspect pioneered by Greene [15] and Katz [17] using Gauss sums, and the p-adic aspect initiated by Dwork [13], in which the p-adic Gamma functions play fundamental roles. The relations between $\Gamma(\cdot)$ and $\Gamma_p(\cdot)$ can be seen from their functional equations; while Gauss sums and $\Gamma_p(\cdot)$ are connected by the Gross–Koblitz formula. In this current series, we first proposed the "Explicit Hypergeometric-Modularity Method" (EHMM) in [1] which provides a direct

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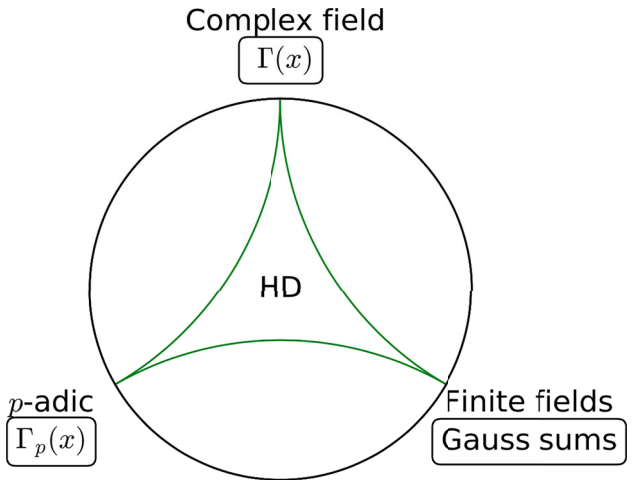


Fig. 1 The three realizations and corresponding fundamental building blocks of hypergeometric motives

link between hypergeometric character sums and coefficients of Hecke eigenforms. The method is based on the Euler integral formula for classical hypergeometric functions, Ramanujan's theory of elliptic functions to alternative bases, Dwork's unit root theory, and the Gross–Koblitz formula connecting the finite field and p-adic settings.

We now recall some basic notation. A *hypergeometric datum* $HD = \{\boldsymbol{\alpha}, \boldsymbol{\beta}\}$ of length n, or written as $\left\{\begin{smallmatrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{smallmatrix}\right\}$, is a pair of multisets $\boldsymbol{\alpha} = \{r_1, r_2, \ldots, r_n\}$ and $\boldsymbol{\beta} = \{q_1, q_2, \ldots, q_n\}$ of rational numbers. Here we assume $q_1 = 1$. Let $F(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda) = {}_nF_{n-1}\left[\begin{matrix} r_1 & r_2 & \ldots & r_n \\ & q_2 & \cdots & q_n \end{matrix}; \lambda\right]$. When $q_1 = 1$ for ${}_nF_{n-1}$, it will be omitted following the classical notation. In [1], we focus on length three or four data at $\lambda = 1$. In this paper, we consider $\lambda = -1$ and length four hypergeometric data as follows: for $1 \le j \le 11$, we let

$$HD_4\left(\frac{j}{12}\right) := \left\{\begin{matrix} \frac{j}{12} & \frac{j}{12} & \frac{1}{2} & \frac{1}{2} \\ 1 & 1 & \frac{1}{2}+\frac{j}{12} & \frac{1}{2}+\frac{j}{12} \end{matrix}\right\}$$

$$HD_5\left(\frac{j}{12}\right) := \left\{\begin{matrix} \frac{j}{12} & 1+\frac{j}{24} & \frac{j}{12} & \frac{1}{2} & \frac{1}{2} \\ 1 & \frac{j}{24} & 1 & \frac{1}{2}+\frac{j}{12} & \frac{1}{2}+\frac{j}{12} \end{matrix}\right\}.$$

Further, we let D denote the least common denominator of $HD_5(j/12)$, and define M to be the least common denominator of D and M. That is,

$$D = 24/\gcd(j, 24) \qquad M = \operatorname{lcm}(4, D). \tag{1.1}$$

Restricting our discussion to primes which are 1 modulo M will simplify the statements of our main results below. The data HD_4 and HD_5 are *well-poised*, meaning that $r_i + q_i$ is independent of i. Moreover HD_5, obtained by inserting a second column with $r_2 = 1+r_1/2$ to $HD_4(j/12)$, is called *very well-poised*, see an article by Andrews for well-poised series [2]. Very well-poised series are of interest in their own right, see for example [44] by Zudilin. Observe that HD_4 is *primitive*—meaning that $r_i - q_j \notin \mathbb{Z}$ for each i, j—while HD_5 is not.

Whipple in [41] studied the consequential symmetries at $\lambda = \pm 1$. Here we combine the EHMM and Whipple's results to study the splitting of the local system associated with $HD_4(j/12)$ at $\lambda = -1$. The splitting behavior leads us to two families of Hecke modular forms. In particular, for $(r, s) \in \mathbb{Q}^2$ we obtain two families of modular forms of weight 2 and 3 respectively from [1,34], defined by

$$\mathbb{K}_1(r,s)(\tau) := \frac{\eta\left(\frac{\tau}{2}\right)^{16s-8r-16}\eta(2\tau)^{8r+8s-12}}{\eta(\tau)^{24s-32}},$$
$$\mathbb{K}_2(r,s)(\tau) := \frac{\eta\left(\frac{\tau}{2}\right)^{16s-8r-12}\eta(2\tau)^{8s+8r-12}}{\eta(\tau)^{24s-30}}.$$

Here η is the Dedekind η function

$$\eta(\tau) = \eta(q) = q^{1/24}\prod_{n\geq 1}(1-q^n), \text{ where } q = e^{2\pi i\tau}, \tau \in \mathbb{C}, \text{ Im}(\tau) > 0.$$

For a modular form f, let $L(f, k)$ denote the L-value of f at k (cf. (3.7)). Our first result, which relates hypergeometric evaluations and explicit special L-values, is as follows:

Theorem 1.1 *For $1 \leq j \leq 11$, let $r = j/12$. Then*

$$\Omega_{r,\mathbb{C}} \cdot F\left(HD_4(r); -1\right) = L\left(\mathbb{K}_1\left(\frac{1-r}{2}, 1+\frac{r}{2}\right)(\tau), 1\right) L\left(\mathbb{K}_2\left(1-\frac{r}{2}, 1+\frac{r}{2}\right)(\tau), 1\right),$$
$$\Omega_{r,\mathbb{C}} \cdot F\left(HD_5(r); -1\right) = -L\left(\mathbb{K}_1\left(\frac{-r}{2}, \frac{3+r}{2}\right)(\tau), 1\right) L\left(\mathbb{K}_2\left(\frac{1-r}{2}, \frac{1+r}{2}\right)(\tau), 1\right),$$

where $\Omega_{r,\mathbb{C}} = 2^{2-6r}\dfrac{\pi}{\sin(r\pi)}\dfrac{\Gamma(r)^2}{\Gamma(r+1/2)^2}$ and $\Gamma(\cdot)$ is the usual Γ function. We also have the following Legendre relation ([7, §1.6] or [36,38]):

$$L\left(\mathbb{K}_1\left(\frac{1-r}{2}, 1+\frac{r}{2}\right)(\tau), 1\right) \cdot L\left(\mathbb{K}_1\left(\frac{-r}{2}, \frac{3+r}{2}\right)(\tau), 1\right) = -\frac{2^{1+6r}}{r\sin(\pi r)}\pi. \tag{1.2}$$

The $\mathbb{K}_1$ and $\mathbb{K}_2$ functions in Theorem 1.1 are not Hecke eigenforms in general, but can be combined to obtain eigenforms.

Theorem 1.2 ([1,33,34]) *Let notation be as above. A suitable combination of $\mathbb{K}_2\left(1-\frac{r}{2}, 1+\frac{r}{2}\right)$ (resp. $\mathbb{K}_1\left(\frac{1-r}{2}, 1+\frac{r}{2}\right)$) and their images under the Hecke operators gives rise to a normalized modular form $f^{\sharp}_{3,D}$ (resp. $f^{\sharp}_{2,D}$), depending on D, which is an eigenform for all Hecke operators T_p with $p \geq 5$. In particular, $f^{\sharp}_{2,D}$ is a CM modular form, see Table 3. The choices of $f^{\sharp}_{2,D}$ and $f^{\sharp}_{3,D}$ are unique up to twisting.*

In Dwork's theory [13], one considers the truncation $F(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda)_{p-1}$ of $F(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda)$ after the pth term. We use the p-adic Gamma functions Γ_p, discussed further in §5 below. For a normalized Hecke eigenform f, we use $a_n(f)$ to denote its nth coefficient and ρ_f for the corresponding Deligne representation of the absolute Galois group $G_{\mathbb{Q}}$ of $\mathbb{Q}$. In particular, because $f^{\sharp}_{2,D}$ is CM, $\rho_{f^{\sharp}_{2,D}}$ decomposes as the sum of two characters when restricted to $G(M)$ for $G(M) := \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\zeta_M))$, with ζ_M a primitive M^{th} root of unity. The p-adic description of this splitting is given in the following theorem:

Table 1 The cuspforms $f^{\sharp}_{2,D}$ and $f^{\sharp}_{3,D}$ denoted using LMFDB labels [22] and with CM information

j	$D(j)$	$M(j)$	$f^{\sharp}_{2,D}$	CM	$f^{\sharp}_{3,D}$	CM
1, 5, 7, 11	24	24	$f_{1152.2.d.g}$	$\mathbb{Q}(\sqrt{-6})$	$f_{1152.3.b.i}$	No
2, 10	12	12	$f_{288.2.a.a}$	$\mathbb{Q}(\sqrt{-1})$	$f_{288.3.g.a}, f_{288.3.g.c}$	No
3, 9	8	8	$f_{128.2.b.a}$	$\mathbb{Q}(\sqrt{-2})$	$f_{128.3.d.c}$	No
4	6	12	$f_{36.2.a.a}$	$\mathbb{Q}(\sqrt{-3})$	$f_{36.3.d.a}, f_{36.3.d.b}$	$\mathbb{Q}(\sqrt{-1})$
6	4	4	$f_{32.2.a.a}$	$\mathbb{Q}(\sqrt{-1})$	$f_{32.3.c.a}$	No
8	3	12	$f_{36.2.a.a}$	$\mathbb{Q}(\sqrt{-3})$	$f_{36.3.d.a}, f_{36.3.d.b}$	$\mathbb{Q}(\sqrt{-1})$

Theorem 1.3 *Let $1 \le j \le 6$, let $r = j/12$, and let D and M be defined as in (1.1). For any prime $p \equiv 1 \pmod{M}$, there exist $\left(a_p(f^{\sharp}_{2,D})\right)_0$ and $\left(a_p(f^{\sharp}_{2,D})\right)_1$ in $\overline{\mathbb{Q}} \cap \overline{\mathbb{Q}}_p$ of p-adic orders 0 and 1, respectively, such that*

$$a_p(f^{\sharp}_{2,D}) = \left(a_p(f^{\sharp}_{2,D})\right)_0 + \left(a_p(f^{\sharp}_{2,D})\right)_1, \quad \text{and} \quad \left(a_p(f^{\sharp}_{2,D})\right)_0 \cdot \left(a_p(f^{\sharp}_{2,D})\right)_1 = p.$$

Moreover, if we let $\Omega_{r,\mathbb{Q}_p} = \dfrac{\Gamma_p(r)^2}{\Gamma_p(r+1/2)^2}$, then

$$\Omega_{r,\mathbb{Q}_p} F(HD_4(r); -1)_{p-1} \equiv \left(a_p(f^{\sharp}_{2,D})\right)_0 a_p(f^{\sharp}_{3,D}) \pmod{p^2}; \tag{1.3}$$

$$\Omega_{r,\mathbb{Q}_p} F(HD_5(r); -1)_{p-1} \equiv \left(a_p(f^{\sharp}_{2,D})\right)_1 a_p(f^{\sharp}_{3,D}) \pmod{p^2}. \tag{1.4}$$

Finally, for the ℓ-adic aspect, we let $H_q(HD; \lambda; \mathfrak{p})$ be the finite hypergeometric function (see (2.4)).

Theorem 1.4 *Let $1 \le j \le 11$, let $r = j/12$, and let D and M be defined as in (1.1). Additionally, let $f^{\sharp}_{2,D}$ and $f^{\sharp}_{3,D}$ be as in Theorem 1.2. For each prime ideal $\mathfrak{p}$ of $\mathbb{Z}[\zeta_M]$ above $p \equiv 1 \pmod{M}$, we have*

$$\Omega_{r,\mathbb{F}_p} H_p\left(HD_4(r); -1; \mathfrak{p}\right) = a_p(f^{\sharp}_{2,D}) \cdot a_p(f^{\sharp}_{3,D}), \tag{1.5}$$

where $\Omega_{r,\mathbb{F}_p} = p\dfrac{\mathfrak{g}(\iota_{\mathfrak{p}}(r))^2}{\mathfrak{g}(\iota_{\mathfrak{p}}(r+1/2))^2}$, $\mathfrak{g}(\cdot)$ denotes the Gauss sum, and $\iota_{\mathfrak{p}}$ is given by (2.3).

Equivalently, if we let $\rho_{\{HD_4(r);-1\}}$ be an explicit hypergeometric representation of $G(M) := \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\zeta_M))$ (see Theorem 2.3) then $\rho_{\{HD_4(r);-1\}}$ is automorphic. In particular,

$$\rho_{\{HD_4(r);-1\}} \simeq \left(\rho_{f^{\sharp}_{2,D}} \otimes \rho_{f^{\sharp}_{3,D}}\right)|_{G(M)} = \left[\left(\chi_{D,1} \otimes \rho_{f^{\sharp}_{3,D}}\right) \oplus \left(\chi_{D,2} \otimes \rho_{f^{\sharp}_{3,D}}\right)\right]|_{G(M)}, \tag{1.6}$$

where $\rho_{f^{\sharp}_{2,D}}|_{G(M)} \simeq \chi_{D,1} \oplus \chi_{D,2}$.

Remark 1.1 As Γ, Γ_p, and $\mathfrak{g}$ are analogs of one another in the complex, p-adic, and finite field settings, respectively, we note that the normalizing terms $\Omega_{r,\mathbb{C}}$, $\Omega_{r,\mathbb{Q}_p}$, and $\Omega_{r,\mathbb{F}_p}$ are similarly analogous to one another.

Using [14], one can realize the corresponding hypergeometric representation from the third étale cohomology group of the smooth model of the following three-fold

$$y^{12/d} = (-1)^d (x_1x_2)^{6/d} (x_3(1-x_1)(1-x_2))^{(12-j)/d} ((1-x_3)(1+x_1x_2x_3))^{j/d}, \tag{1.7}$$

where $d = \gcd(j, 6)$. The well-poised assumption on $HD_4(j/12)$ at $\lambda = -1$ can be realized through the following involution acting on (1.7):

$$\mathfrak{i} : \begin{cases} x_i & \mapsto 1/x_i, \quad i = 1, 2, 3 \\ y & \mapsto (-1)^{j/12} \cdot y/(x_1x_2x_3)^2. \end{cases} \tag{1.8}$$

This involution leads to the final decomposition in (1.6). Over finite fields of cardinality $q \equiv 1 \pmod{M}$ the values in (1.5) correspond to point counts of this variety over $\mathbb{F}_q$ by [14, Proposition 4.2]. Further discussions related to the compactifications of such families can be found in recent work [19].

The $j = 6$ case of Theorem 1.4 was obtained in [27] by McCarthy–Papanikolas. To demonstrate our results, we now summarize a few conclusions after specializing to this case.

Proposition 1.5 *For $j = 6$, $HD_4(1/2) = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1, 1, 1, 1 \right\}$, $f_{3,6}^\sharp = f_{32.3.c.a} = \mathbb{K}_2\left(\frac{1}{4}, \frac{3}{4}\right)(8\tau) + 4i\mathbb{K}_2\left(\frac{3}{4}, \frac{5}{4}\right)(8\tau)$ as listed in Table 2, $f_{2,6}^\sharp = f_{32.2.a.a} = \mathbb{K}_1(\frac{1}{4}, \frac{5}{4})(8\tau)$ as in Table 3.*

- *Over $\mathbb{C}$:*

$$ {}_4F_3\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ & 1 & 1 & 1 \end{matrix}; -1\right] = \frac{32}{\pi^2} \cdot L(f_{32.2.a.a}, 1) \cdot |\operatorname{Im}(L(f_{32.3.c.a}, 1))|,$$

$$ {}_5F_4\left[\begin{matrix} \frac{1}{2} & \frac{5}{4} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ & \frac{1}{4} & 1 & 1 & 1 \end{matrix}; -1\right] = \frac{2}{\pi} \cdot L(f_{32.2.a.a}, 1)^{-1} \cdot \operatorname{Re}(L(f_{32.3.c.a}, 1)),$$

where $L(f_{32.2.a.a}, 1) = \frac{1}{8} \frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}\right)}$; and

$$L\left(f_{32.3.c.a}, 1\right) = \frac{\Gamma\left(\frac{1}{2}\right)}{8} \left(\frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)} {}_3F_2\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{4} \\ & 1 & \frac{3}{4} \end{matrix}; 1\right] + 4\sqrt{-1} \frac{\Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}\right)} {}_3F_2\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & \frac{3}{4} \\ & 1 & \frac{5}{4} \end{matrix}; 1\right] \right).$$

- *Over $\mathbb{F}_p$: For each prime $p > 3$,*

$$H_p\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & 1 & 1 & 1 \end{matrix}; -1\right] = a_p(f_{32.2.a.a}) a_p(f_{32.3.c.a}).$$

In this case, the L-function $L(HD_4(1/2); -1; s)$[1] is the Rankin–Selberg convolution of $L(f_{32.2.a.a}, s)$ and $L(f_{32.3.c.a}, s)$.

[1] It can be computed by a `Magma` package implemented by Watkins [39] (cf. also [10] by H. Cohen).

- *Over $\mathbb{Q}_p$: For each prime $p \equiv 1 \pmod 4$,*

$$a_p\left(f_{32.2.a.a}\right) = \left(a_p\left(f_{32.2.a.a}\right)\right)_0 + p/\left(a_p\left(f_{32.2.a.a}\right)\right)_0, \text{ and}$$
$$\left(a_p\left(f_{32.2.a.a}\right)\right)_0 = \frac{\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)}{\Gamma_p\left(\frac{3}{4}\right)}.$$

Then

$$\begin{aligned} {}_4F_3\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ & 1 & 1 & 1\end{matrix}; -1\right]_{p-1} &\equiv p \cdot {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{3}{4} \\ & 1 & \frac{5}{4}\end{matrix}; 1\right]_{p-1} \\ &\equiv \left(a_p(f_{32.2.a.a})\right)_0 a_p(f_{32.3.c.a}) \pmod{p^2}. \end{aligned}$$

$$\begin{aligned} {}_5F_4\left[\begin{matrix}\frac{1}{2} & \frac{5}{4} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ & \frac{1}{4} & 1 & 1 & 1\end{matrix}; -1\right]_{p-1} &\equiv p \cdot {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{4} \\ & 1 & \frac{3}{4}\end{matrix}; 1\right]_{p-1} \\ &\equiv \frac{p}{\left(a_p(f_{32.2.a.a})\right)_0} a_p(f_{32.3.c.a}) \pmod{p^2}. \end{aligned}$$

This paper is organized as follows. We first provide some background in Sect. 2, in Sect. 3 we first recall the EHMM method, then the $\mathbb{K}_1$ and $\mathbb{K}_2$-functions and derive their properties. Well-poised formulae will be recalled and used in Sect. 4. Then we prove our main results over $\mathbb{C}$ and $\mathbb{F}_q$. The p-adic aspect is discussed in Sect. 5. In the appendix, we use the hypergeometric values to explore special L-values of the $\mathbb{K}_2$-functions and some of their quadratic twists.

2 Notation and hypergeometric functions

2.1 Notation and classical theory

Here we briefly recall the necessary notation and definitions from Part I of this series. For a more thorough introduction to hypergeometric functions and character sums in our setting, we refer the reader to [1]. As in the introduction, we use *HD* to denote a *hypergeometric datum* $\{\boldsymbol{\alpha}, \boldsymbol{\beta}\}$ or $\left\{\begin{matrix}\boldsymbol{\alpha} \\ \boldsymbol{\beta}\end{matrix}\right\}$, where $\boldsymbol{\alpha} = \{r_1, r_2, \ldots, r_n\}$ and $\boldsymbol{\beta} = \{q_1, q_2, \ldots, q_n\}$ are two multisets of rational numbers. We say that *HD* is *primitive* if $r_i - q_j \notin \mathbb{Z}$ for all $1 \le i, j \le n$, and that *HD* is *defined over* $\mathbb{Q}$ if both

$$\prod_{j=1}^{n} x - e^{2\pi i r_j} \quad \text{and} \quad \prod_{j=1}^{n} x - e^{2\pi i q_j}$$

have integer coefficients. We refer to n as the *length* of *HD*.

A useful aspect of classical hypergeometric functions is the integral representation of Euler. When $\mathrm{Re}(q_i) > \mathrm{Re}(r_i) > 0$, this gives an inductive formula to construct hypergeometric functions, see for example [3, (2.2.2)]. In the following, we assume that $q_1 = 1$. First, define ${}_1P_0[r_1; z] := (1-z)^{-r_1}$. Then for $n \ge 2$ define the (classical) period function as

$${}_nP_{n-1}\left[\begin{matrix} r_1 & r_2 & \cdots & r_n \\ & q_2 & \cdots & q_n \end{matrix}; z\right] = \int_0^1 t^{r_n}(1-t)^{q_n - r_n - 1} {}_{n-1}P_{n-2}\left[\begin{matrix} r_1 & r_2 & \cdots & r_{n-1} \\ & q_2 & \cdots & q_{n-1} \end{matrix}; zt\right] \frac{dt}{t}. \tag{2.1}$$

The classical ${}_nF_{n-1}$ function is related to ${}_nP_{n-1}$ by ${}_1F_0[r_1;z] := {}_1P_0[r_1;z]$, and

$$ {}_nF_{n-1}\left[\begin{matrix} r_1 & r_2 & \cdots & r_n \\ & q_2 & \cdots & q_n \end{matrix}; z\right] = \prod_{i=2}^{n} \frac{\Gamma(q_i)}{\Gamma(r_i)\Gamma(q_i - r_i)} \cdot {}_nP_{n-1}\left[\begin{matrix} r_1 & r_2 & \cdots & r_n \\ & q_2 & \cdots & q_n \end{matrix}; z\right], \tag{2.2} $$

For convenience, in this series of papers, we adopt the convention that

$$ f\left(\frac{a_1^{r_1}, \cdots, a_m^{r_m}}{b_1^{s_1}, \cdots, b_n^{s_n}}\right) := \frac{f(a_1)^{r_1} \cdots f(a_m)^{r_m}}{f(b_1)^{s_1} \cdots f(b_n)^{s_n}} $$

where f could be the Gamma function $\Gamma(\cdot)$, the Pochhammer symbol $(\cdot)_n$, the p-adic Gamma function $\Gamma_p(\cdot)$, or the Gauss sum $\mathfrak{g}(\cdot)$. Similarly, we use $f\left(a_1^{r_1}, \cdots, a_m^{r_m}\right)$ to denote $f(a_1)^{r_1} \cdots f(a_m)^{r_m}$. Finally, for a fixed multiset $\boldsymbol{\alpha} = \{r_1, \ldots, r_n\}$, we use the notation $f(\boldsymbol{\alpha}) := f(r_1, r_2, \ldots, r_n)$.

2.2 Some background over the finite fields

Let $\mathbb{F}_q$ be a finite field of odd characteristic and $\widehat{\mathbb{F}_q^\times} = \langle \omega \rangle$ be the set of multiplicative characters for $\mathbb{F}_q^\times$. We take the convention that $A(0) = 0$ for each $A \in \widehat{\mathbb{F}_q^\times}$. Let ε be the trivial character, ϕ be the quadratic character, and $\bar{A}$ be the inverse of A.

Further, the following shorthand notations for Gauss and Jacobi sums are used. Let $a, b \in \mathbb{Q}$ such that $a(q-1), b(q-1) \in \mathbb{Z}$, then

$$ \mathfrak{g}_\omega(a) := \sum_{x \in \mathbb{F}_q} \omega^{(q-1)a}(x) \zeta_p^{\mathrm{Tr}_{\mathbb{F}_p}^{\mathbb{F}_q}(x)} $$

and

$$ J_\omega(a, b) := \sum_{x \in \mathbb{F}_q} \omega^{(q-1)a}(x) \omega^{(q-1)b}(1-x), $$

where $\mathrm{Tr}_{\mathbb{F}_p}^{\mathbb{F}_q}$ is the standard trace map from $\mathbb{F}_q$ to $\mathbb{F}_p$.

2.2.1 Hypergeometric character sums

In [14], a finite character sum ${}_n\mathbb{P}_{n-1}$ is defined inductively, parallel to (2.1). We now describe how to pass from $\mathbb{Z}[\zeta_M]$ to finite fields. For each nonzero prime ideal $\mathfrak{p}$ coprime to M and integer i we assign a character to the residue field $\kappa_\mathfrak{p} := \mathbb{Z}[\zeta_M]/\mathfrak{p}$ using the Mth residue symbol by

$$ \iota_\mathfrak{p}\left(\frac{i}{M}\right)(x) := \left(\frac{x}{\mathfrak{p}}\right)_M^i \equiv x^{(q-1)\frac{i}{M}} \pmod{\mathfrak{p}}, \quad \forall x \in \mathbb{Z}[\zeta_M]. \tag{2.3} $$

For example, if $\mathfrak{p}$ is coprime to 2, $\iota_\mathfrak{p}(1/2) = \phi_\mathfrak{p}$ or simply ϕ, the quadratic character. Likewise, $\iota_\mathfrak{p}(1)$ is the trivial character, denoted by $\varepsilon_\mathfrak{p}$ or ε. By definition $\iota_\mathfrak{p}(i/M)(0) = 0$ for each i. In this notation, we denote Gauss sums by $\mathfrak{g}(\iota_\mathfrak{p}(r_i))$ and Jacobi sums by $J(\iota_\mathfrak{p}(r_i), \iota_\mathfrak{p}(r_j))$. For simplicity, we write R_i for $\iota_\mathfrak{p}(r_i)$, define Q_i analogously, and use q for $|\kappa_\mathfrak{p}|$. We refer the reader to [4] or [14] for further background on Gauss and Jacobi sums. We now replace our classical hypergeometric functions with character sums using the

dictionary first used by Greene [15] and described explicitly in [14, §2.4]. The first key adjustment is to replace t^{r_i} with $R_i(t)$, where $t \in \mathbb{F}_q$. Note the Pochhammer symbols in the classical setting are quotients of gamma functions. The gamma functions are then replaced with Gauss sums. Let R_i, Q_i, q as above. For a fixed $\lambda \in \mathbb{F}_q^\times$ and $n \geq 1$, define

$$\begin{aligned}\mathbb{P}(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \mathfrak{p}) &= {}_n\mathbb{P}_{n-1}\left[\begin{matrix} R_1 & R_2 & \cdots & R_n \\ Q_1 & Q_2 & \cdots & Q_n \end{matrix}; \lambda; q\right] \\ &:= \frac{(-1)^n}{q-1} \cdot \left(\prod_{i=2}^{n} R_i Q_i(-1)\right) \cdot \sum_{\chi \in \widehat{\mathbb{F}_q^\times}} \binom{R_1\chi}{Q_1\chi}\binom{R_2\chi}{Q_2\chi}\cdots\binom{R_n\chi}{Q_n\chi}\chi(\lambda),\end{aligned}$$

where $\binom{A}{B} := -B(-1)J(A, \overline{B})$ taking values in $\mathbb{Q}(\zeta_M)$ or $\mathbb{Q}_\ell(\zeta_M)$. When the pair $(\boldsymbol{\alpha}, \boldsymbol{\beta})$ is primitive, let

$$H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \mathfrak{p}) := (-1)^{n-1}\mathcal{J}(HD; \mathfrak{p})^{-1}\mathbb{P}(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \mathfrak{p}), \tag{2.4}$$

where

$$\mathcal{J}(HD; \mathfrak{p}) := \iota_{\mathfrak{p}}(r_1 + q_1)(-1)\prod_{i=1}^{n} -J(\iota_{\mathfrak{p}}(r_i), \iota_{\mathfrak{p}}(q_i - r_i)). \tag{2.5}$$

Note that $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \mathfrak{p})$ can be written in terms of Gauss sums as in [5,26] or (4.3) of [1]. If $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \mathfrak{p}) \in \mathbb{Q}$, independent of which prime ideal $\mathfrak{p}$ above $(p) = \mathfrak{p} \cap \mathbb{Z}$, we replace $\mathfrak{p}$ by p as in [1], or simply omit it. In certain applications, especially in §5, we will fix an embedding of $\mathbb{Q}(\zeta_M)$ to $\overline{\mathbb{C}}_p$ by taking $\iota_{\mathfrak{p}}\left(\frac{1}{q-1}\right) := \overline{\omega}_q$ to be the inverse of the Teichmüller character ω_q. Under this embedding, we write $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \mathfrak{p})$ as $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}; \lambda; \overline{\omega}_q)$.

2.3 Hypergeometric Galois representations

To motivate the connection between character sums and Galois representations, we recall a result of Weil which allows us to view Jacobi sums as Grössencharacters. Below, $G(M)$ denotes the absolute Galois group of $\mathbb{Q}(\zeta_M)$ as before.

Theorem 2.1 (Weil [40]) *Let $r, s \in \mathbb{Q}$ such that $r, s, r+s \notin \mathbb{Z}$. Let $M = \mathrm{lcd}(r, s)$. Then there exists a representation χ of $G(M)$ such that at each nonzero prime ideal $\mathfrak{p} \nmid M$ of $\mathbb{Z}[\zeta_M]$,*

$$\chi(Frob_{\mathfrak{p}}) = -J(\iota_{\mathfrak{p}}(r), \iota_{\mathfrak{p}}(s)),$$

in which $Frob_{\mathfrak{p}}$ denotes the geometric Frobenius conjugacy class of $G(M)$ at $\mathfrak{p}$.

The Gross–Koblitz formula acts as a bridge between the finite field and p-adic settings. The formula can be stated in terms of Jacobi sums as follows:

Lemma 2.2 (Corollary 2.7 of [1]) *Let $r, s \in \mathbb{Q} \cap (0, 1)$ and $p \equiv 1 \pmod{\mathrm{lcd}(r, s)}$ such that $r + s < 1$. Then*

$$-J_{\tilde{\omega}_p}(r, s) = \Gamma_p\left(\frac{r, s}{r+s}\right).$$

Next we recall the following result of Katz which puts hypergeometric character sums into the Galois perspective.

Theorem 2.3 (Katz [17,18]) *Let ℓ be a prime. Given a primitive pair of multi-sets $\boldsymbol{\alpha} = \{r_1, \cdots, r_n\}$, $\boldsymbol{\beta} = \{q_1 = 1, q_2, \cdots, q_n\}$ with $M = M(HD)$, for any $\lambda \in \mathbb{Z}[\zeta_M, 1/M] \smallsetminus \{0\}$ the following hold:*

(i). There exists an ℓ-adic Galois representation $\rho_{\{HD;\lambda\}} : G(M) \to GL(W_\lambda)$ unramified almost everywhere such that at each prime ideal $\mathfrak{p}$ of $\mathbb{Z}[\zeta_M, 1/(M\ell\lambda)]$ with residue field $\kappa_\mathfrak{p}$ and norm $q = N(\mathfrak{p}) = |\kappa_\mathfrak{p}|$,

$$\mathrm{Tr}\rho_{\{HD;\lambda\}}(Frob_\mathfrak{p}) = (-1)^{n-1}\iota_\mathfrak{p}(r_1)(-1) \cdot \mathbb{P}(\boldsymbol{\alpha}, \boldsymbol{\beta}; 1/\lambda; \mathfrak{p}).$$

(ii). When $\lambda \neq 1$, the dimension $d := dim_{\overline{\mathbb{Q}}_\ell} W_\lambda$ equals n. All roots of the characteristic polynomial of $\rho_{\{HD;\lambda\}}(Frob_\mathfrak{p})$ are algebraic numbers and have the same absolute value $N(\mathfrak{p})^{(n-1)/2}$ under all archimedean embeddings.

(iii). When $\lambda = 1$, $d = n - 1$.

3 The $\mathbb{K}_2$-functions and the proof of Theorem 1.2

3.1 The explicit hypergeometric-modularity method

In the first paper of this series [1] we establish various congruences between the Fourier coefficients of newforms, truncated classical hypergeometric functions, and finite field hypergeometric functions to establish the modularity of certain hypergeometric Galois representations. For the reader's convenience, we begin by restating the main result of the first paper.

Theorem 3.1 ([1]) *Let $n \in \{3, 4\}$. Assume $\boldsymbol{\alpha}^\flat = \{r_1, \cdots, r_{n-1}\}$, where $0 < r_1 \leq \cdots \leq r_{n-1} < 1$, and $\boldsymbol{\beta}^\flat = \{1, \cdots, 1\}$, with r_n, q_n such that $0 < r_n < q_n \leq 1$ and $r_2 < q_n$. Let $HD = \{\{r_n\} \cup \boldsymbol{\alpha}^\flat, \{q_n\} \cup \boldsymbol{\beta}^\flat\}$ and $M = M(HD)$. Further assume $\gamma(HD) := n - 2 + q_n - \sum_{i=1}^n r_i \geq 1$. Suppose HD satisfies the following hypotheses:*

1. *There exists a modular function $t = C_1 q + O(q^2) \in \mathbb{Z}[[q]]$ such that*

$$f_{HD}(q) := C_1^{-r_n} \cdot t(q)^{r_n}(1 - t(q))^{q_n - r_n - 1} F(\boldsymbol{\alpha}^\flat, \boldsymbol{\beta}^\flat; t(q)) \frac{dt(q)}{t(q)dq} \tag{3.1}$$

   is a congruence weight n holomorphic cuspform satisfying that, for each prime $p \equiv 1 \pmod{M}$, $T_p f_{HD} = \tilde{b}_p f_{HD}$ for some $\tilde{b}_p$ in $\mathbb{Z}$ where T_p is the pth Hecke operator.
2. *For any prime ideal $\mathfrak{p}$ in $\mathbb{Z}[\zeta_M]$ above $p \equiv 1 \pmod{M}$ we have*

$$\chi_{HD}(\mathfrak{p}) \cdot \mathbb{P}(HD; 1; \mathfrak{p}) \in \mathbb{Z}, \tag{3.2}$$

   where

$$\chi_{HD}(\mathfrak{p}) := \iota_\mathfrak{p}(r_n)(C_1)^{-1} \cdot \prod_{i=2}^{n-1} \iota_\mathfrak{p}(r_i)(-1). \tag{3.3}$$

Then there exists a normalized Hecke eigenform $f_{HD}^{\sharp}$, not necessarily unique, such that for each $p \equiv 1 \pmod{M}$, $\tilde{b}_p = a_p(f_{HD}^{\sharp})$, and at each prime $p \geq 29$ congruent to 1 modulo M

$$(-1)^{n-1}\chi_{HD}(\mathfrak{p}) \cdot \mathbb{P}(HD; 1; \mathfrak{p}) = a_p(f_{HD}^{\sharp}) + \delta_{\gamma(HD)=1} \cdot \psi_{HD}(p) \cdot p. \tag{3.4}$$

Here $\delta_{\gamma(HD)=1}$ is equal to 1 when $\gamma(HD) = 1$ and is 0 otherwise and $\psi_{HD} = \pm 1$ satisfying that

$$\psi_{HD}(p) \equiv (-1)^{n-1} \cdot \iota_p(r_n)(C_1)\Gamma_p\left(\frac{q_n - r_n}{\boldsymbol{\alpha}^{\flat}}\right) \pmod{p}.$$

3.2 The $\mathbb{K}_2$-functions revisited

Throughout this paper, $n = 3$, $\boldsymbol{\alpha}^{\flat} = \{\frac{1}{2}, \frac{1}{2}\}$, $\boldsymbol{\beta}^{\flat} = \{1, 1\}$, and so we suppress the subscripts on r_n and q_n. To avoid confusion between the parameter q_3 and $q = e^{2\pi i\tau}$ we use s for q_3 in this paper. Further, let t be the modular lambda function

$$\lambda(\tau) = 16\frac{\eta\left(\frac{\tau}{2}\right)^8 \eta(2\tau)^{16}}{\eta(\tau)^{24}}.$$

Lemma 3.2 ([1]) *Given $r, s \in \mathbb{Q}$,*

$$\lambda(\tau)^r(1-\lambda(\tau))^{s-r-1}\,{}_2F_1\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} \\ 1\end{matrix}; \lambda(\tau)\right]\frac{d\lambda(\tau)}{\lambda(\tau)dq} = 2^{4r-1}\mathbb{K}_2(r,s)(\tau), \tag{3.5}$$

where, as in the introduction,

$$\mathbb{K}_2(r,s)(\tau) := \frac{\eta\left(\frac{\tau}{2}\right)^{16s-8r-12}\eta(2\tau)^{8s+8r-12}}{\eta(\tau)^{24s-30}}.$$

Further, we recall a particular set of parameters by

$$\mathbb{S}_2 := \left\{(r,s) \mid 0 < r < s < 3/2, r \neq 1, s \neq \frac{1}{2},\ 24s \in \mathbb{Z}, 8(r+s) \in \mathbb{Z}\right\}.$$

The set $\mathbb{S}_2$ is chosen such that for each (r, s) in $\mathbb{S}_2$, $\mathbb{K}_2(r, s)$ is a congruence holomorphic cuspidal modular form of weight three, and the datum $\left\{\left\{\frac{1}{2}, \frac{1}{2}, r\right\}, \{1, 1, s\}\right\}$ is primitive.

Definition 3.1 ([1]) Given $r, s \in \mathbb{Q}$, define

$$\mathbb{K}_2^*(r,s)(\tau) := \mathbb{K}_2(s-r,s)(\tau). \tag{3.6}$$

For $r \in \mathbb{Q}_{\geq 0}$ such that $24r \in \mathbb{Z}$ we define

$$N(r) := \frac{48}{\gcd(24r, 24)},$$

which is also 2 times the denominator of r when written in reduced form.

Lemma 3.3 (Lemma 3.1 [1]) *For each $(r, s) \in \mathbb{S}_2$, $\mathbb{K}_2(r, s)(N(r)\tau)$ is a congruence weight three holomorphic cusp form of level $N(r)N(s-r)$ with Dirichlet character $\left(\frac{-2^{24s}}{\cdot}\right)$. Moreover, $\mathbb{K}_2(r,s)(-1/\tau) = 2^{4s-8r}i \cdot \tau^3\mathbb{K}_2^*(r,s)(\tau)$.*

3.3 The Galois and non-Galois cases

Returning to Theorem 3.1, we note that for each $(r, s) \in \mathbb{S}_2$ the hypothesis (1) is satisfied by the function $\mathbb{K}_2(r, s)$. The next definition addresses when hypothesis (2), namely (3.2), holds for $HD = \{\{\frac{1}{2}, \frac{1}{2}, r\}, \{1, 1, s\}\}$. For a given $(r, s) \in \mathbb{S}_2$ and $c \in \mathbb{Z}_{>0}$, let $(r_c, s_c) \equiv c(r, s)$ mod $\mathbb{Z}$ be the corresponding conjugate pair in $\mathbb{S}_2$. That is, let r_c be the fractional part of cr, which we denote by $\{cr\}$, and let $s_c = \{cs\}$ if $\{cr\} - \{cs\} \leq 0$ and $s_c = \{cs\} + 1$ otherwise.

Definition 3.2 A pair $(r, s) \in \mathbb{S}_2$ is said to be in a *Galois orbit* for the $\mathbb{K}_2$-family if for any $c \in (\mathbb{Z}/N\mathbb{Z})^\times$, $\mathbb{K}_2(r, s)$ and $\mathbb{K}_2(r_c, s_c)$ are in the same Hecke orbit. In this case, we use $f^\sharp_{HD}$ to denote a normalized Hecke eigenform in the orbit.

Let O be the Galois orbit associated to HD. Then $f^\sharp_{HD} = \sum_{(r,s)\in O} h(r, s)\mathbb{K}(r, s)$ where $h(r, s) \in \overline{\mathbb{Q}}$. There are non-Galois cases, which are equally interesting, such as $\{\left(\frac{1}{8}, \frac{1}{4}\right), \left(\frac{5}{8}, \frac{5}{4}\right), \left(\frac{3}{8}, \frac{3}{4}\right)\}$. These pairs are conjugates in $\mathbb{S}_2$. The first two are in the same Hecke orbit, for example,

$$\mathbb{K}_2\,(1/8, 1/4)\,(16\tau) - 8\mathbb{K}_2\,(5/8, 5/4)\,(16\tau) = f_{256.3.c.a},$$
$$\mathbb{K}_2\,(1/8, 1/4)\,(16\tau) + 8\mathbb{K}_2\,(5/8, 5/4)\,(16\tau) = f_{256.3.c.b},$$

while $\mathbb{K}_2\left(\frac{3}{8}, \frac{3}{4}\right)(16\tau)$ is in the orbit of $f_{256.3.c.d}$. See [1, Section 3.2] for the determination of $h(r, s)$ and the work by Rosen [33] for more detailed discussion.

We recall the explicit modularity result for the Galois families here.

Theorem 3.4 [1, Theorem 1.1] *Let $HD := \{\{\frac{1}{2}, \frac{1}{2}, r\}, \{1, 1, s\}\}$, $M = M(HD)$, and $\rho_{\{HD;1\}} : G(M) \longrightarrow GL_2(\overline{\mathbb{Q}}_\ell)$ be the associated Katz representation (cf Theorem 2.3). If $(r, s) \in \mathbb{S}_2$ is in a Galois orbit O, then there is an explicit finite character χ_r of $G(M)$ depending on r such that*

$$\left(\chi_r \otimes \rho_{\{HD;1\}}\right)|_{G(2M)} \simeq \rho_{f^\sharp_{HD}}|_{G(2M)},$$

where $\rho_{f^\sharp_{HD}}$ denotes the Deligne representation of $G_{\mathbb{Q}}$ associated with a Hecke eigenform $f^\sharp_{HD}$.

We will focus on the following family in this paper:

$$\text{G2}: \left(\frac{j}{24}, \frac{24-j}{24}\right) \text{ for } j \in \{1, \cdots, 11\}.$$

Let $D = 24/\gcd(j, 24)$. For each D, the corresponding orbit is

$$O(D) = \left\{\left(\frac{i}{D}, \frac{D-i}{D} + \left\lfloor\frac{2i}{D}\right\rfloor\right) \mid i \in (\mathbb{Z}/D\mathbb{Z})^\times\right\}.$$

In Table 2, we give the forms $f^\sharp_{\{\frac{1}{2},\frac{1}{2},\frac{j}{24}\},\{1,1,\frac{24-j}{24}\}}$ for the G2 family. As it only depends on D not j, we denote the form by $f^\sharp_{3,D}$.

3.4 Special L-values of the $\mathbb{K}_2(r, s)$ functions

Below, we will relate the previous discussion to special L-values. For a given weight-k modular form, recall that

$$L(f, 1) := -2\pi i \int_0^{i\infty} f(t)dt, \quad L(f, 2) := -4\pi^2 \int_0^{i\infty} f(t)tdt. \tag{3.7}$$

Also recall when $s > r$,

$$ {}_3P_2\left[\begin{matrix} r_1 & r_2 & r \\ & q_2 & s \end{matrix}; z\right] = \int_0^1 t^{r-1}(1-t)^{s-r-1}\, {}_2P_1\left[\begin{matrix} r_1 & r_2 \\ & q_2 \end{matrix}; tz\right] dt. $$

Consequently, one has the following lemma:

Lemma 3.5 *Assume the notation as in Lemma 3.2, and let $N = N(r)$. Then*

$$ \begin{aligned} {}_3P_2\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & r \\ 1 & & s \end{matrix}; 1\right] &= 2^{4r-1}N\pi \cdot L\left(\mathbb{K}_2(r,s)(N\tau), 1\right) \\ &\overset{(3.6)}{=} 2^{4s-4r-2}N^2 \cdot L\left(\mathbb{K}_2(s-r,s)(N\tau), 2\right). \end{aligned} $$

Proof Recall that the modular lambda function $\lambda(\tau)$ satisfies that $\lambda(0) = 1$ and $\lambda(i\infty) = 0$. We see that

$$ \begin{aligned} {}_3P_2\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & r \\ 1 & & s \end{matrix}; 1\right] &= \pi \int_0^1 \lambda^r(1-\lambda)^{s-r-1}\, {}_2F_1\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} \\ & 1 \end{matrix}; \lambda\right] \frac{d\lambda}{\lambda} \\ &\overset{(3.5)}{=} -\pi^2 i \cdot 2^{4r} \cdot \int_0^{i\infty} \mathbb{K}_2(r,s)(\tau) d\tau \\ &= \pi 2^{4r-1} \cdot L\left(\mathbb{K}_2(r,s)(\tau), 1\right) \\ &\overset{\tau\mapsto -1/\tau}{=} 2^{4s-4r-2} \cdot L\left(\mathbb{K}_2(s-r,s)(\tau), 2\right), \end{aligned} $$

in the last step we apply the identity $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$. Our claims follow from the substitution $\tau \mapsto N\tau$. □

3.5 The $\mathbb{K}_1$-functions

The idea behind Theorem 3.1 can be used more generally. For example, when $\boldsymbol{\alpha} = \{\frac{1}{2}\}$, $\boldsymbol{\beta} = \{1\}$, for $(r,s) \in \mathbb{Q}^2$ and $t = \lambda(\tau)$, Equation (3.1) gives rise to a weight two modular form

$$ \mathbb{K}_1(r,s)(\tau) = 2^{1-4r}\left(\frac{\lambda}{1-\lambda}\right)^r (1-\lambda)^{s-1/2}\theta_3^4(\tau), \tag{3.8} $$

for some finite index subgroup of $\mathrm{SL}_2(\mathbb{Z})$. These $\mathbb{K}_1(r,s)(\tau)$ functions have also been considered by Rosen [34]. Using an argument similar to Lemma 3.5, one has

$$ L(\mathbb{K}_1(r,s)(\tau), 1) = 2^{1-4r}B(r, s-r-1/2). \tag{3.9} $$

We list the Hecke eigenforms of Theorem 1.2 in Tables 2 and 3.

Table 2 The $f^{\sharp}_{3,D}$ functions arising from the G2 class

D	N	Hecke Eigenform $f^{\sharp}_{3,D}$ in terms of $\mathbb{K}_2(r,s)(N\tau)$
24	48	$f_{1152.3.b.i} = \mathbb{K}_2\left(\frac{1}{24},\frac{23}{24}\right) - \beta_1\mathbb{K}_2\left(\frac{5}{24},\frac{19}{24}\right) - \beta_5\mathbb{K}_2\left(\frac{7}{24},\frac{17}{24}\right) - \beta_3\mathbb{K}_2\left(\frac{11}{24},\frac{13}{24}\right)$ $-\beta_2\mathbb{K}_2\left(\frac{13}{24},\frac{35}{24}\right) - \beta_4\mathbb{K}_2\left(\frac{17}{24},\frac{31}{24}\right) - \beta_7\mathbb{K}_2\left(\frac{19}{24},\frac{29}{24}\right) - \beta_6\mathbb{K}_2\left(\frac{23}{24},\frac{25}{24}\right)$
12	24	$f_{288.3.g.a} = \mathbb{K}_2\left(\frac{1}{12},\frac{11}{12}\right) - 4\mathbb{K}_2\left(\frac{5}{12},\frac{7}{12}\right) + 4i\mathbb{K}_2\left(\frac{7}{12},\frac{17}{12}\right) - 16i\mathbb{K}_2\left(\frac{11}{12},\frac{13}{12}\right)$ $f_{288.3.g.c} = f_{288.3.g.a}\otimes\chi_3, \qquad \chi_3 = \left(\frac{3}{\cdot}\right)$
8	16	$f_{128.3.d.c} = \mathbb{K}_2\left(\frac{1}{8},\frac{7}{8}\right) + 2\sqrt{2}\mathbb{K}_2\left(\frac{3}{8},\frac{5}{8}\right) + 4i\mathbb{K}_2\left(\frac{5}{8},\frac{11}{8}\right) + 8\sqrt{2}i\mathbb{K}_2\left(\frac{7}{8},\frac{9}{8}\right)$
6	12	$f_{36.3.d.a}(\tau) + 2f_{36.3.d.a}(2\tau) = \mathbb{K}_2\left(\frac{1}{6},\frac{5}{6}\right)(12\tau) + 8\mathbb{K}_2\left(\frac{5}{6},\frac{7}{6}\right)(12\tau)$ $f_{36.3.d.b}(\tau) - 2f_{36.3.d.b}(2\tau) = \mathbb{K}_2\left(\frac{1}{6},\frac{5}{6}\right)(12\tau) - 8\mathbb{K}_2\left(\frac{5}{6},\frac{7}{6}\right)(12\tau)$
4	8	$f_{32.3.c.a} = \mathbb{K}_2\left(\frac{1}{4},\frac{3}{4}\right) + 4i\mathbb{K}_2\left(\frac{3}{4},\frac{5}{4}\right)$
3	6	$f_{36.3.d.a} = \mathbb{K}_2\left(\frac{1}{3},\frac{2}{3}\right) - 2\mathbb{K}_2\left(\frac{2}{3},\frac{4}{3}\right)$ $f_{36.3.d.b} = f_{36.3.d.a}\otimes\chi_{-3} = f_{36.3.d.a}\otimes\chi_3, \qquad \chi_{-3} = \left(\frac{-3}{\cdot}\right),\ \chi_3 = \left(\frac{3}{\cdot}\right)$

Here $\beta_1 = 2i\sqrt{7}$, $\beta_2 = 4i$, $\beta_3 = 4\sqrt{2}$, $\beta_4 = 8\sqrt{7}$, $\beta_5 = 2i\sqrt{14}$, $\beta_6 = -16i\sqrt{2}$, $\beta_7 = 8\sqrt{14}$, and $\mathbb{K}_2(r,s)(\tau) = \frac{\eta(\tau/2)^{16s-8r-12}\eta(2\tau)^{8s+8r-12}}{\eta(\tau)^{24s-30}}$

Table 3 The $f^{\sharp}_{2,D}$ functions

D	N	Hecke Eigenform $f^{\sharp}_{2,D}$ in terms of $\mathbb{K}_1(r,s)(N\tau)$	CM
24	48	$f_{1152.2.d.g} = \mathbb{K}_1(\frac{1}{24},\frac{35}{24}) + 2\sqrt{3}i\,\mathbb{K}_1(\frac{5}{24},\frac{31}{24})$ $+2\sqrt{6}\,\mathbb{K}_1(\frac{7}{24},\frac{29}{24}) - 4\sqrt{2}i\,\mathbb{K}_1(\frac{11}{24},\frac{25}{24})$	-24
12	24	$f_{288.2.a.a} = \mathbb{K}_1(\frac{1}{12},\frac{17}{12}) - 4\,\mathbb{K}_1(\frac{5}{12},\frac{13}{12})$	-4
8	16	$f_{128.2.b.a} = \mathbb{K}_1(\frac{1}{8},\frac{11}{8}) + 2\sqrt{2}i\,\mathbb{K}_1(\frac{3}{8},\frac{9}{8})$	-8
6	6	$f_{36.2.a.a} = \mathbb{K}_1(\frac{1}{3},\frac{7}{6})$	-3
4	8	$f_{32.2.a.a} = \mathbb{K}_1(\frac{1}{4},\frac{5}{4})$	-4
3	12	$f_{36.2.a.a} = \mathbb{K}_1(\frac{1}{6},\frac{4}{3})$	-3

$\mathbb{K}_1(r,s)(\tau) = \frac{\eta(\tau/2)^{16s-8r-16}\eta(2\tau)^{8r+8s-12}}{\eta(\tau)^{24s-32}}$

4 Four dimensional hypergeometric Galois representations

4.1 Some well-poised formulae over $\mathbb{C}$ and $\mathbb{F}_q$

Given a hypergeometric system in variable λ, the variable change $\lambda \mapsto 1/\lambda$ will lead to another hypergeometric system playing a "dual" role. See for example [6] by Beukers–Jouhet. Note that well-poised hypergeometric local systems here are self-dual and hence the involution

$$\mathfrak{i} : \lambda \mapsto 1/\lambda$$

fixing $\lambda = \pm 1$ plays an important role in the hypergeometric arithmetic (cf. Proposition 2 of [21]). For the cases in this paper, it can be realized by the map (1.8) acting on (1.7). In [41], Whipple investigated well-poised series and very well-posed series at ± 1 and derived a few reduction formulae. For this paper, putting two of them together gives a more complete picture of Theorem 1.1. To make the notation compact, we use $r_i^* = 1 + r_1 - r_i$.

Proposition 4.1 (Whipple) *Let $r_i \in \mathbb{C}$. Whenever both sides converge,*

$$\begin{aligned} {}_4F_3\left[\begin{matrix} r_1 & r_2 & r_3 & r_4 \\ & r_2^* & r_3^* & r_4^* \end{matrix}; -1\right] &= C(r_i)\cdot {}_3F_2\left[\begin{matrix} 1+\frac{r_1}{2}-r_2 & r_3 & r_4 \\ & 1+\frac{r_1}{2} & r_2^* \end{matrix}; 1\right] \\ {}_5F_4\left[\begin{matrix} r_1 & 1+\frac{r_1}{2} & r_2 & r_3 & r_4 \\ & \frac{r_1}{2} & r_2^* & r_3^* & r_4^* \end{matrix}; -1\right] &= C(r_i)\cdot {}_3F_2\left[\begin{matrix} \frac{1+r_1}{2}-r_2 & r_3 & r_4 \\ & \frac{1+r_1}{2} & r_2^* \end{matrix}; 1\right], \end{aligned} \tag{4.1}$$

where

$$C(r_i) = \frac{\Gamma(r_3^*)\Gamma(r_4^*)}{\Gamma(1+r_1)\Gamma(r_4^* - r_3)}.$$

Note that the second datum $\left\{\begin{matrix} r_1, 1+\frac{r_1}{2}, r_2, r_3, r_4 \\ 1, \ \frac{r_1}{2}, \ r_2^*, r_3^*, r_4^* \end{matrix}\right\}$ on the left-hand side is very well-poised.

Proof The first one is (3.4) of [41]. For the second one, we use (6.3) of [41], which states

$${}_6F_5\left[\begin{matrix} r_1 & 1+\frac{r_1}{2} & r_5 & r_2 & r_3 & r_4 \\ & \frac{r_1}{2} & r_5^* & r_2^* & r_3^* & r_4^* \end{matrix}; -1\right] = \frac{\Gamma(r_3^*)\Gamma(r_4^*)}{\Gamma(1+r_1)\Gamma(r_4^* - r_3)}\, {}_3F_2\left[\begin{matrix} r_5^* - r_2 & r_3 & r_4 \\ & r_5^* & r_2^* \end{matrix}; 1\right]. \tag{4.2}$$

When $r_5 = \frac{1+r_1}{2}$, $r_5^* = r_5$ and hence the ${}_6F_5(-1)$ becomes the desired ${}_5F_4(-1)$. □

In comparison, we recall a well-poised formula of McCarthy over finite fields. The notation $\chi = \square$ is used to indicate the character χ is a square.

Theorem 4.2 (McCarthy [26]) *Let $r_1, r_2, r_3, r_4 \in \mathbb{Q}$ such that all of the following data are primitive and M is the least positive common denominator of the r_i's. Let $q \equiv 1 \pmod{M}$ be an odd prime power. Then the following H_q values are 0 if $\omega^{(q-1)r_1} \neq \square$ and when $\omega^{(q-1)r_1} = \square$,*

$$H_q\left(\begin{matrix} r_1 & r_2 & r_3 & r_4 \\ 1 & r_2^* & r_3^* & r_4^* \end{matrix}; -1\right) = \mathfrak{g}_\omega\left(\frac{-r_1, r_3 - r_4^*}{-r_3^*, -r_4^*}\right) \sum_{s \in \left\{\frac{r_1}{2}, \frac{r_1+1}{2}\right\}} H_q\left(\begin{matrix} s - r_2 & r_3 & r_4 \\ 1 & s & r_2^* \end{matrix}; 1\right). \tag{4.3}$$

In this formula, the right hand side contains two summands. Comparing (4.1) and (4.3) closely, we see that not only $\frac{\Gamma(r_3^*)\Gamma(r_4^*)}{\Gamma(1+r_1)\Gamma(r_4^*-r_3)}$ and $\mathfrak{g}_\omega\left(\frac{-r_1, r_3-r_4^*}{-r_3^*, -r_4^*}\right)$ match up to reflection formulae, but also the two choices of s correspond to the two ${}_3F_2(1)$'s on the right-hand sides of (4.1) respectively.

4.2 Length four data relating to $\mathbb{K}_2$-functions

Letting

$$r_1 = r_2 = r, \quad r_3 = r_4 = \frac{1}{2}, \tag{4.4}$$

on the right-hand side of Proposition 4.1 corresponds to the L-values of $\mathbb{K}_2\left(1 - \frac{r}{2}, 1 + \frac{r}{2}\right)$ and $\mathbb{K}_2\left(\frac{1-r}{2}, \frac{1+r}{2}\right)$ at 1, respectively. In this case

$$C(r) = \frac{1}{r}\Gamma\left(\frac{r + \frac{1}{2}, r + \frac{1}{2}}{r, r}\right). \tag{4.5}$$

Remark 4.1 Like the discussion in [11], the local system of $HD_4(r)$ at $\lambda = -1$ consists of differential 3-forms of the first and second kind on the variety (1.7). When $\lambda = -1$, these differentials admit the induced action of $\mathfrak{i}$ given by (1.8). For example, for $\omega = \frac{dx_1 dx_2 dx_3}{y}$ then $\mathfrak{i}^*\omega = u\omega$, for some 48th root of unity u, while for $\tilde{\omega} = (x_1 - 1/x_1)(x_2 - 1/x_2)(x_3 - 1/x_3)\omega$, $\mathfrak{i}^*\tilde{\omega} = -u\tilde{\omega}$. Under the specialization (4.4), the $C(r)$ factor can be absorbed using the P-function notation, see the discussion below.

4.3 Proof of Theorem 1.1

Lemma 4.3 *For any real r,*

$$B\left(\frac{1-r}{2}, r\right) B\left(-\frac{r}{2}, 1+r\right) = -\frac{2^{1+2r}\pi}{r\sin(\pi r)}. \tag{4.6}$$

when both hand sides are defined.

Proof The standard properties of gamma and beta functions from [14, Section 2] give

$$\begin{aligned} B\left(\frac{1-r}{2}, r\right) B\left(-\frac{r}{2}, 1+r\right) &= \frac{\Gamma\left(\frac{1-r}{2}\right)\Gamma(r)\Gamma(-\frac{r}{2})\Gamma(1+r)}{\Gamma\left(\frac{1+r}{2}\right)\Gamma\left(1+\frac{r}{2}\right)} \\ &= 2^{1+2r}\frac{\Gamma(r)}{\Gamma(1+r)}\cdot\Gamma(-r)\Gamma(1+r) = -\frac{2^{1+2r}\pi}{r\sin(\pi r)}. \end{aligned}$$

□

We are now prepared to prove Theorem 1.1.

Proof of Theorem 1.1 Recall $\mathbb{K}_1(r,q)(\tau)$ from (3.8). When $0 < r < 1/2$, $\mathbb{K}_1\left(\frac{1-r}{2}, 1+\frac{r}{2}\right)(\tau)$ is a holomorphic weight two cuspform while $\mathbb{K}_1\left(\frac{-r}{2}, \frac{3+r}{2}\right)(\tau)$ is not holomorphic. By Equation (2.2)

$$\begin{aligned} P(HD_4(r);-1) &= B(r,1-r)B(1/2,r)^2\cdot F(HD_4(r);-1) \\ &\overset{(4.1)}{=} \frac{B(r,1-r)B(1/2,r)^2\cdot C(r)}{B(1/2,1/2)}\frac{1}{B(1-r/2,r)}\,{}_3P_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & 1-\frac{r}{2} \\ & 1+\frac{r}{2} & 1\end{matrix}; 1\right]. \end{aligned}$$

Applying (4.5), (4.6), and properties of the Gamma and Beta functions, we get

$$P(HD_4(r);-1) = \frac{1}{2^{2r}}B\left(\frac{1-r}{2}, r\right)\cdot {}_3P_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & 1-\frac{r}{2} \\ 1 & 1 & 1+\frac{r}{2}\end{matrix}; 1\right].$$

By Eq. (3.9) and Lemma 3.5, this agrees with

$$2^{6r-2}\pi\cdot L\left(\mathbb{K}_1\left(\frac{1-r}{2}, 1+\frac{r}{2}\right)(\tau), 1\right)\cdot L\left(\mathbb{K}_2\left(1-\frac{r}{2}, 1+\frac{r}{2}\right)(\tau), 1\right).$$

Similarly,

$$\begin{aligned} &\left(1+\frac{2}{r}\cdot z\frac{d}{dz}\right){}_4P_3\left[\begin{matrix}r & r & \frac{1}{2} & \frac{1}{2} \\ 1 & r+\frac{1}{2} & r+\frac{1}{2}\end{matrix}; z\right]\Bigg|_{z=-1} \\ &\quad = B(r,1-r)B(r,1/2)^2\;{}_5F_4\left[\begin{matrix}r & 1+\frac{r}{2} & r & \frac{1}{2} & \frac{1}{2} \\ & \frac{r}{2} & 1 & r+\frac{1}{2} & r+\frac{1}{2}\end{matrix}; -1\right]. \end{aligned}$$

Thus by (4.1), the left hand side equals

$$\begin{aligned} &-\frac{1}{2^{2r}}B\left(\frac{-r}{2}, r\right)\cdot {}_3P_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1-r}{2} \\ & 1 & \frac{1+r}{2}\end{matrix}; 1\right] \\ &\quad = -2^{6r-2}\pi\cdot L\left(\mathbb{K}_1\left(\frac{-r}{2}, \frac{3+r}{2}\right)(\tau), 1\right)\cdot L\left(\mathbb{K}_2\left(\frac{1-r}{2}, \frac{1+r}{2}\right)(\tau), 1\right). \end{aligned}$$

At last we prove Equation 1.2. By Equation 3.9

$$\begin{aligned} &L\left(\mathbb{K}_1\left(\frac{1-r}{2}, 1+\frac{r}{2}\right)(\tau), 1\right) L\left(\mathbb{K}_1\left(\frac{-r}{2}, \frac{3+r}{2}\right)(\tau), 1\right) \\ &= 2^{1-2(1-r)} B((1-r)/2, r) 2^{1+2r} B(-r/2, 1+r) \\ &\overset{(4.6)}{=} -\frac{2^{1+6r}\pi}{r\sin(\pi r)} \end{aligned}$$

□

4.4 Proof of Theorem 1.4

The finite field analogue of Lemma 4.3 is as follows:

Lemma 4.4 *Let $r \in \mathbb{Q}$ and D be the denominator of $r/2$ and $q \equiv 1 \pmod{D}$ be an odd prime power. Then*

$$J_\omega\left(\frac{1-r}{2}, r\right) J_\omega\left(1-\frac{r}{2}, r\right) = q\,\omega^{(q-1)r}(4). \tag{4.7}$$

Proof The standard properties of Gauss and Jacobi sums from [14, Section 2] give

$$J_\omega\left(\frac{1-r}{2}, r\right) J_\omega\left(1-\frac{r}{2}, r\right) = \omega^{(q-1)r}(4)\frac{\mathfrak{g}_\omega(1/2)\mathfrak{g}_\omega(-r)\mathfrak{g}_\omega(r)^2}{\mathfrak{g}_\omega(1/2)\,\mathfrak{g}_\omega(r)} = q\,\omega^{(q-1)r}(4).$$

□

We now use the Jacobi sums in the previous Lemma to describe the pth coefficients of the $f^\sharp_{2,D}$ functions, as listed in Table 3.

Lemma 4.5 *Let $j \in \{1, 2, 3, 4, 6\}$, $D = 24/j$, $r = j/12$, and $M = \mathrm{lcm}(4, D)$. For any prime $p \equiv 1 \pmod{M}$,*

$$\begin{aligned} a_p\left(f^\sharp_{2,D}\right) &= -\left(\mathbb{P}\left(\left\{\frac{1}{2}, \frac{r}{2}\right\}, \left\{1, \frac{1-r}{2}\right\}; 1; \mathfrak{p}\right) + \mathbb{P}\left(\left\{\frac{1}{2}, \frac{r+1}{2}\right\}, \left\{1, \frac{-r}{2}\right\}; 1; \mathfrak{p}\right)\right) \\ &\quad \cdot \iota_\mathfrak{p}(r)\left(\frac{1}{4}\right) \\ &= -\left[J_\omega\left(\frac{r}{2}, -r\right) + J_\omega\left(\frac{r+1}{2}, -r\right)\right]\omega^{(1-p)r}(4). \end{aligned} \tag{4.8}$$

This means $\rho_{f^\sharp_{2,D}}|_{G(M)}$ is isomorphic to the direct sum of two characters of $G(M)$.

Before proving Lemma 4.5, we first mention a corollary. As $a_p\left(f^\sharp_{2,D}\right) \in \mathbb{Z}$, it is independent of the choice of $\omega \in \widehat{\mathbb{F}_p^\times}$ and how we embed ω into $\mathbb{Z}_p$. Taking the embedding to be $\tilde{\omega}_p$, by Corollary 2.2 we obtain a description of the p-adic unit root of $a_p\left(f^\sharp_{2,D}\right)$, to be used in the proof of Proposition 5.4.

Corollary 4.6 *With notation as in the previous Lemma and $p \equiv 1 \pmod{D}$,*

$$a_p\left(f^\sharp_{2,D}\right) = \left(a_p\left(f^\sharp_{2,D}\right)\right)_0 + p/\left(a_p\left(f^\sharp_{2,D}\right)\right)_0,$$

where

$$\left(a_p\left(f_{2,D}^{\sharp}\right)\right)_0 = \Gamma_p\left(\frac{\frac{r}{2}, 1-r}{1-\frac{r}{2}}\right)\bar{\omega}_p^{(1-p)r}(4) = -(-1)^{(p-1)\frac{r}{2}}\bar{\omega}_p(4)^{(p-1)r}\cdot\frac{\Gamma_p(r)}{\Gamma_p\left(\frac{1+r}{2}\right)^2}. \tag{4.9}$$

Here is an outline of the next proof. Using Theorem 2.1, we regard the sum of the two Jacobi sums as a 2-dimensional representation of $G(M)$ (which is extendable to $G_{\mathbb{Q}}$). Then we compare it with $\rho_{f_{2,D}^{\sharp}}|_{G(M)}$.

Proof of Lemma 4.5 Here we assume $\iota_{\mathfrak{p}}\left(\frac{1}{p-1}\right) = \bar{\omega}_p$, see (2.3) for the notation.

First, we explain the relation between the two $\mathbb{P}$-functions in Equation (4.8). By the finite field Gauss summation formula ([14, (6.11)]),

$$u_{D,p} := -\mathbb{P}\left(\left\{\frac{1}{2}, \frac{r}{2}\right\}, \left\{1, \frac{1-r}{2}\right\}; 1; \mathfrak{p}\right)\iota_{\mathfrak{p}}(r)\left(\frac{1}{4}\right) = -J_{\bar{\omega}_p}\left(\frac{r}{2}, -r\right)\bar{\omega}_p^{(1-p)r}(4).$$

By Theorem 2.1, this can be viewed as the Frobenius trace of a character, denoted by $\chi_{D,1}$, of $G(M)$. Note that $|u_{D,p}| = \sqrt{p}$ under any archimedean embedding. A similar conclusion can be drawn for the second summand, which we will denote by $v_{D,p}$ and use $\chi_{D,2}$ to denote the corresponding character of $G(M)$. It is not isomorphic to $\chi_{D,1}$. Let $\tau : \zeta_M \mapsto \zeta_M^{M/2+1} \in \mathrm{Gal}(\mathbb{Q}(\zeta_M)/\mathbb{Q})$. It is straightforward to check that τ swaps $\chi_{D,1}$ and $\chi_{D,2}$.

Next, we show that the claim holds modulo p. From Table 3, when $p \equiv 1 \pmod{D}$

$$a_p\left(f_{2,D}^{\sharp}\right) = a_p\left(\mathbb{K}_1\left(\frac{r}{2}, \frac{1-r}{2}\right)(2D\tau)\right) = a_p\left(\mathbb{K}_1\left(\frac{1}{D}, \frac{3}{2}-\frac{1}{D}\right)(2D\tau)\right)$$

is an integer bounded by $2\sqrt{p}$. We expand the function $(\lambda/16)^{1/D}(1-\lambda)^{-2/D}$ arising from $\mathbb{K}_1(1/D, 3/2-1/D)(\tau)$ as a power series of $\lambda^{1/D}$. Using properties of commutative formal group laws as in [1, §3.3], its pth coefficient, $4^{-\frac{2}{D}}\frac{\Gamma(2/D+(p-1)/D)}{\Gamma(2/D)\Gamma(1+(p-1)/D)} \equiv a_p\left(f_{2,D}^{\sharp}\right) \pmod{p}$. Thus for $p \equiv 1 \pmod{M}$

$$\begin{aligned} a_p\left(f_{2,D}^{\sharp}\right) &\equiv 4^{-2/D}\frac{\Gamma(2/D+(p-1)/D)}{\Gamma(2/D)\Gamma(1+(p-1)/D)} \\ &\overset{(5.4)}{\equiv} \Gamma_p\left(\frac{\frac{1}{D}, 1-\frac{2}{D}}{1-\frac{1}{D}}\right)4^{2(p-1)/D} \pmod{p} \\ &\equiv -J_{\bar{\omega}_p}(1/D, 1-2/D)\,4^{2(p-1)/D} \pmod{p} \\ &\equiv u_{D,p}+v_{D,p} \pmod{p} \end{aligned}$$

as by the Gross–Koblitz formula, $v_{D,p} = -J_{\bar{\omega}_p}\left(\frac{1}{D}+\frac{1}{2}, 1-\frac{2}{D}\right)4^{2(p-1)/D} \in p\mathbb{Z}_p$. If the right hand side of (4.8) is also an integer, the claim holds when $p > 5$. To prove this claim, we will apply the idea used in the proof of Proposition 6.1 of [1]. Pick an integral basis $b_1, \cdots, b_m$ for $\mathbb{Z}[\zeta_M]$ as a free-$\mathbb{Z}$-module. So $w_{D,p} := u_{D,p}+v_{D,p}-a_p\left(f_{2,D}^{\sharp}\right) \in \mathbb{Z}[\zeta_M]$ can be written as $\sum_{i=1}^m a_i b_i$ where $a_i \in \mathbb{Z}$. For each $\sigma_k : \zeta_M \mapsto \zeta_M^k$ where $k \in (\mathbb{Z}/M\mathbb{Z})^{\times}$, $\sigma_k(u_{D,p}+v_{D,p})$ consists of two other $\mathbb{P}$-functions with j replaced by kj. For each $1 \le i \le D/2$ coprime to D, $\mathbb{K}_1(i/D, 3/2-i/D)(2D\tau)$ is in the same Hecke orbit as $\mathbb{K}_1(1/D, 3/2-1/D)(2D\tau)$, so

$\sigma_k(w_{D,p}) \equiv 0 \pmod{p}$ for each k as above. Thus $p \mid a_i$ for each $1 \le i \le m$. Meanwhile, under each archimedean embedding of $\mathbb{Q}(\zeta_M)$ the norm of $w_{D,p}$ is less than $4\sqrt{p}$. Thus for $p > 17$, each $a_i = 0$. Namely $w_{D,p} = 0$. Smaller primes can be verified directly.

At last, to verify the p-adic expressions for $\left(a_p\left(f^{\sharp}_{2,D}\right)\right)_0$, we use the reflection formula (5.2) and the duplication formula for Γ_p, see [9]. Using these gives

$$\begin{aligned}\Gamma_p\left(\frac{\frac{r}{2}, 1-r}{1-\frac{r}{2}}\right)\bar{\omega}_p^{(1-p)r}(4) &= \Gamma_p\left(\frac{(r+1)/2, \frac{r}{2}, 1-r, r, 1/2}{1-\frac{r}{2}, r, r/2, (r+1)/2, (r+1)/2}\right)\\ &= \Gamma_p\left(\frac{r, 1/2, 1-r, r, 1/2}{1-\frac{r}{2}, r, r/2, (r+1)/2, (r+1)/2}\right)\bar{\omega}_p(1/4)^{(1-p)r}\\ &= -(-1)^{(p-1)r/2}\bar{\omega}_p(4)^{(p-1)3r}\cdot\frac{\Gamma_p(r)}{\Gamma_p\left(\frac{1+r}{2}\right)^2}.\end{aligned} \tag{4.10}$$

□

We are now ready to prove Theorem 1.4 in a manner parallel to the proof of Theorem 1.1.

Proof of Theorem 1.4 Set

$$HD_3(u, v) := \left\{\left\{\frac{1}{2}, \frac{1}{2}, u\right\}; \{1, 1, v\}\right\}. \tag{4.11}$$

When $r_3 = r_4 = \frac{1}{2}, r_1 = r_2 = r$ (in which case $r_2^* = 1$), the datum on the right hand side of (4.3) is $HD_3(s-r, s)$. Recall that $r = j/12$, and $D = 24/\gcd(j, 24)$ is double the denominator of r in the lowest case. We let $M = \mathrm{lcm}(4, D)$ so that, for any $\mathfrak{p}$ above $p \equiv 1 \pmod{M}$, $-J(\iota_{\mathfrak{p}}(r), \iota_{\mathfrak{p}}(1-r)) = 1$ and $-J(\iota_{\mathfrak{p}}(\frac{1}{2}), \iota_{\mathfrak{p}}(\frac{1}{2})) = 1$. Thus, by Theorem 2.3, for any prime ideal $\mathfrak{p}$ above $p \equiv 1 \pmod{M}$

$$\mathrm{Tr}\,\rho_{\{HD_4(r);-1\}}(\mathrm{Frob}_{\mathfrak{p}}) = -\mathbb{P}(HD_4(r); -1; \mathfrak{p}).$$

When $s = \frac{r}{2}$, or $\frac{r+1}{2}$, by Theorem 1.2 and Theorem 3.1,

$$\mathbb{P}(HD_3(s-r, s); 1, \mathfrak{p}) \cdot \iota_{\mathfrak{p}}(s-r)(1/16) = a_p(f^{\sharp}_{3,D}). \tag{4.12}$$

Using the conversion between the $\mathbb{P}$ and H_p functions in (2.4), we get

$$-\mathbb{P}(HD_4(r); -1; \mathfrak{p}) \overset{(2.4)}{=} J\left(\iota_{\mathfrak{p}}(r), \iota_{\mathfrak{p}}\left(\frac{1}{2}\right)\right)^2 H_p(HD_4(r); -1; \mathfrak{p}) \tag{4.13}$$

Next we apply McCarthy's well-poised theorem, Theorem 4.2, to get

$$
\begin{aligned}
-\mathbb{P}(HD_4(r);-1;\mathfrak{p}) &= p\cdot \sum_{s\in\{\frac{r}{2},\frac{r+1}{2}\}} H_p(HD_3(s-r,s);1;\mathfrak{p}) \\
&= -\sum_{s\in\{\frac{r}{2},\frac{r+1}{2}\}} \frac{p}{J\left(\iota_{\mathfrak{p}}(r),\iota_{\mathfrak{p}}(-s)\right)}\mathbb{P}\left(HD_3\left(s-r,s\right);1;\mathfrak{p}\right).
\end{aligned}
$$

Finally, we finish the conclusion using the expressions for $a_p(f^{\#}_{i,D})$ for $i=2,3$

$$
\begin{aligned}
&-\mathbb{P}(HD_4(r);-1;\mathfrak{p}) \\
&\overset{(4.12)}{=} a_p\left(f^{\#}_{3,D}\right)\iota_{\mathfrak{p}}(r)(1/4)\left[-\frac{p}{J\left(\iota_{\mathfrak{p}}(r),\iota_{\mathfrak{p}}\left(-\frac{r}{2}\right)\right)}-\frac{p}{J\left(\iota_{\mathfrak{p}}(r),\iota_{\mathfrak{p}}\left(\frac{1-r}{2}\right)\right)}\right] \\
&\overset{(4.7)}{=} a_p\left(f^{\#}_{3,D}\right)\cdot\iota_{\mathfrak{p}}(r)(1/4)^2\left[-J\left(\iota_{\mathfrak{p}}(r),\iota_{\mathfrak{p}}\left(\frac{1-r}{2}\right)\right)-J\left(\iota_{\mathfrak{p}}(r),\iota_{\mathfrak{p}}\left(\frac{-r}{2}\right)\right)\right] \\
&\overset{(4.8)}{=} a_p\left(f^{\#}_{3,D}\right)a_p\left(f^{\#}_{2,D}\right).
\end{aligned}
$$

The result now follows from (4.13) and equation (2.13) of [14], which implies

$$
\Omega_{r,\mathbb{F}_p} = J\left(\iota_{\mathfrak{p}}(r),\iota_{\mathfrak{p}}\left(\frac{1}{2}\right)\right)^2 = p\frac{\mathfrak{g}(\iota_{\mathfrak{p}}(r))^2}{\mathfrak{g}(\iota_{\mathfrak{p}}(r+1/2))^2}.
$$

In terms of representations, this means $\rho_{\{HD(r);-1\}} \simeq \left(\rho_{f^{\sharp}_{2,D}}\otimes\rho_{f^{\sharp}_{3,D}}\right)|_{G(M)}$. □

We now consider two cases with data defined over $\mathbb{Q}$. By the work of Beukers, Cohen, and Mellit [5]; the H_p functions can be extended to almost all primes p for such data.

Corollary 4.7 *Consider the hypergeometric data $HD_4(1/2)$, defined in §1, and $HD_1 = \left\{\left\{\frac{1}{6},\frac{1}{6},\frac{5}{6},\frac{5}{6}\right\},\left\{\frac{1}{3},\frac{1}{3},\frac{2}{3},\frac{2}{3}\right\}\right\}$. Then for all odd primes we have*

$$
H_p\left(HD_4(1/2);-1\right) = a_p(f_{32.3.c.a})a_p(f_{32.2.a.a}),
$$

and for all primes $p>3$

$$
p^2H_p\left(HD_1;-1\right) = a_p(f_{288.3.g.c})a_p(f_{288.2.a.a}).
$$

Equivalently, if we let $\tilde{\rho}_{\{HD_4(1/2);-1\}}$ and $\tilde{\rho}_{\{HD_1;-1\}}$ be representations of $G_{\mathbb{Q}}$ such that at unramified primes p,

$$
\begin{aligned}
&\mathrm{Tr}\tilde{\rho}_{\{HD_4(1/2);-1\}}(Frob_p) \\
&\quad = H_p(HD_4(1/2);-1) \text{ and } \mathrm{Tr}\tilde{\rho}_{\{HD_1;-1\}}(Frob_p) = p^2H_p(HD_1;-1).
\end{aligned}
$$

Then

$$
\tilde{\rho}_{\{HD_4(1/2);-1\}} \cong \rho_{f_{32.3.c.a}}\otimes\rho_{f_{32.2.a.a}}
$$

and

$$
\tilde{\rho}_{\{HD_1;-1\}} \cong \rho_{f_{288.3.g.c}}\otimes\rho_{f_{288.2.a.a}} \cong \rho_{f_{288.3.g.a}}\otimes\rho_{f_{288.2.a.e}}.
$$

Proof For $HD_4(1/2)$, by Theorem 1.4, we only need to consider primes $p \equiv 3 \pmod 4$. By Theorem 4.2, Then $H_p(HD_4(1/2)1;-1) = 0 = a_p(f_{32.2.a.a})$, at primes $p \equiv 3 \pmod 4$. The vanishing of $a_p(f_{32.2.a.a})$ follows since $f_{32.2.a.a}$ has CM by $\mathbb{Q}(\sqrt{-1})$.

Now $H_p(HD_1;-1) = 0$ at primes $p \equiv 3 \pmod 4$ since

$$\begin{aligned} H_p(HD_1;-1) =& \frac{1}{1-p}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} G(\chi)^2\chi(-1) \overset{\chi\mapsto\phi\overline{\chi}}{=} \frac{1}{1-p}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} G(\phi\overline{\chi})^2\phi\overline{\chi}(-1) \\ =& \phi(-1)H_p(HD_1;-1), \end{aligned}$$

where $G(\chi) = \overline{\chi}(2^4)\dfrac{\mathfrak{g}(\chi^6)\mathfrak{g}(\chi)\mathfrak{g}(\overline{\chi}^3)}{\mathfrak{g}(\chi^2)\mathfrak{g}(\chi^3)\mathfrak{g}(\overline{\chi})} = \dfrac{\mathfrak{g}(\phi\chi^3)\mathfrak{g}(\overline{\chi}^3)}{\mathfrak{g}(\phi\chi)\mathfrak{g}(\overline{\chi})}$, by [14, Theorem 2.7]. Then, for $p \equiv 3 \pmod 4$, $H_p(HD_1;-1) = 0 = a_p(f_{288.2.a.a}) = a_p(f_{288.2.a.c})$, since the weight-two forms have CM by $\mathbb{Q}(\sqrt{-1})$.

To complete our proof, it suffices to check primes $p \equiv 1 \pmod{12}$ together with an additional prime which is 5 modulo 12, by the Chebotarev density theorem. Below we use $p = 5$.

Fix a generator ω of $\widehat{\mathbb{F}_p^\times}$. The substitution $\chi \mapsto \chi\omega^{\frac{p-1}{3}}$ and Theorem 4.2 give

$$\begin{aligned} p^2\cdot H_p(HD_2;-1) =& J_\omega(1/3,1/6)^2\cdot H_p(HD_4(1/6);-1) \\ =& -\frac{J_\omega(1/3,1/6)^2}{J_\omega(1/2,1/6)^2}\mathbb{P}(HD_4(1/6);-1;p) = -\mathbb{P}(HD_4(1/6),-1;p) \\ =& -(J_\omega(5/6,1/12)+J_w(5/6,7/12))\,\overline{\omega}^{\frac{p-1}{3}}(2)a_p(f_{3,12}^\sharp). \end{aligned}$$

The multiplication formula for Gauss sums [14, Theorem 2.7] and Lemma 4.5 give

$$\begin{aligned} &-(J_w(5/6,1/12)+J_w(5/6,7/12))\,\overline{\omega}^{\frac{p-1}{3}}(2) \\ &= -\overline{\omega}^{\frac{p-1}{4}}(-3)J_w(1/2,1/4) - \omega^{\frac{p-1}{4}}(-3)J_w(1/2,3/4) = a_p(f_{288.2.a.a}). \end{aligned}$$

Now [1, Theorem 1.1] gives $a_p(f_{3,12}^\sharp) = a_p(f_{288.3.g.c}) = a_p(f_{288.3.g.a})$ for $p \equiv 1 \pmod{12}$. To decide the exact combination, we note that $f_{288.3.g.c} = f_{288.3.g.a}\otimes\chi_3$ and use $p = 5$ to determine the forms by comparing

$$\begin{aligned} 5^2\cdot H_5(HD;-1) = -16 &= 4a_5(f_{288.2.a.a}) = a_5(f_{288.3.g.c})a_5(f_{288.2.a.a}) \\ &= -4a_5(f_{288.2.a.e}) = a_5(f_{288.3.g.a})a_5(f_{288.2.a.e}). \end{aligned}$$

This gives the results. □

5 *p*-Adic aspects and the proof of Theorem 1.3

5.1 Some *p*-adic background and the *p*-adic perturbation method

For the p-adic perspective, we introduce Morita's [28] p-adic Γ_p function.

Definition 5.1 The p-adic Gamma function $\Gamma_p : \mathbb{Z}_p \to \mathbb{Z}_p^\times$ is defined on positive integers n by

$$\Gamma_p(n) := (-1)^n \prod_{\substack{1\le i<n \\ p\nmid i}} i,$$

and then extended continuously to $\mathbb{Z}_p$.

This function satisfies properties similar to the classical Γ function, which can be quickly checked for positive integers and then extended by continuity to all of $\mathbb{Z}_p$. First, we have an analog to the functional equation of the Γ function

$$\frac{\Gamma_p(x+1)}{\Gamma_p(x)} = \begin{cases} -x & \text{if } x \in \mathbb{Z}_p^\times \\ -1 & \text{if } x \in p\mathbb{Z}_p. \end{cases} \tag{5.1}$$

As an analog to Euler's reflection formula, we have

$$\Gamma_p(x)\Gamma_p(1-x) = (-1)^{x_0}, \tag{5.2}$$

where x_0 is the unique solution to $x \equiv x_0 \pmod{p}$ with $1 \leq x_0 \leq p$. We will need to translate between classical Γ functions, Pochhammer symbols, and p-adic Γ_p functions. To do so, we observe that $\Gamma_p(n)$ is nearly $(n-1)!$ for $n \in \mathbb{Z}^+$, differing only by a sign and the missing multiples of p. In particular,

$$(n-1)! = (-1)^n \Gamma_p(n) \prod_{j=1}^{\lfloor \frac{n-1}{p} \rfloor} jp.$$

This can then be extended continuously to general Pochhammer symbols, which allows us to rewrite quotients of Γ-functions as quotients of Γ_p-functions when the inputs differ by an integer by first expressing the quotient as a Pochhammer symbol. See Lemma 17 of [24] for more details. Doing this gives us the following:

Lemma 5.1 *Let $r \in \mathbb{Q} \cap \mathbb{Z}_p$, and let $0 \leq k \leq p-1$. We have*

$$(r)_k = (-1)^k \frac{\Gamma_p(r+k)}{\Gamma_p(r)} (r + [-r]_0)^{\nu(k,[-r]_0)} \tag{5.3}$$

where $[\cdot]_0$ denotes the first p-adic digit and $\nu(k,x)$ is defined to be 0 if $k \leq x$ and 1 if $k > x$. Further, if $a - b \in \mathbb{Z}$, then

$$\frac{\Gamma(a)}{\Gamma(b)} = (-1)^{a-b} \frac{\Gamma_p(a)}{\Gamma_p(b)} v_{a,b,p}, \tag{5.4}$$

where $v_{a,b,p}$ is the product of those numbers x between a and b for which $x - a \in \mathbb{Z}$ and $x \in p\mathbb{Z}_p$, or the reciprocals thereof in the case $b > a$.

We now consider the analytic properties of Γ_p. Define functions

$$G_i(x) := \frac{\frac{d^i}{dx^i}\Gamma_p(x)}{\Gamma_p(x)}.$$

The following result of the third author and Ramakrishna allows us to approximate quotients of p-adic Γ-functions in terms of these G_i functions.

Theorem 5.2 (Theorem 14, [24]) *Suppose $p \geq 5$, $r \in \mathbb{N}$, $a \in \mathbb{Z}_p$, $m \in \mathbb{C}_p$ satisfies $v_p(m) \geq 0$ and $t \in \{0, 1, 2\}$. Then*

$$\frac{\Gamma_p(a + mp^r)}{\Gamma_p(a)} \equiv \sum_{k=0}^{t} \frac{G_k(a)}{k!}(mp^r)^k \pmod{p^{(t+1)r}}.$$

The above result holds for $t = 4$ if $p \geq 11$.

This result is the foundation of the *p-adic perturbation* method. The key idea of this method is to perturb the parameters of a hypergeometric datum by adding terms which are divisible by p. When these perturbations are chosen suitably, they can allow for better manipulations of the hypergeometric functions when viewed over the complex numbers or finite fields, while only creating small variations p-adically by Theorem 5.2. In the previous part of this series, we used rational functions to generate hypergeometric identities suitable to our purposes. In this paper, we are able to exploit pre-existing hypergeometric identities, significantly simplifying the calculations.

Before moving to the supercongruences, we begin with an application of Theorem 5.2 to remove perturbations from quotients of Γ_p-functions modulo high powers of p by averaging over certain sums.

Proposition 5.3 *Let $\ell \leq m$ be integers, p be a prime, and let $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_\ell, \ldots \xi_m) \subset \mathbb{Z}_p^m$. Define*

$$C_\ell(\boldsymbol{\xi}) := \frac{\prod_{i=1}^{\ell} \Gamma_p(\xi_i)}{\prod_{i=\ell+1}^{m} \Gamma_p(\xi_i)}.$$

Fix vectors $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_m) \in \mathbb{Z}_p^m$ and $\mathbf{w} = (w_1, \ldots, w_n) \in \mathbb{Z}_p^n$ with $\mathbf{w}$ satisfying $\sum_{i=1}^{n} w_i = 0$ and $p \nmid n$. Then

$$\frac{1}{n} \sum_{j=1}^{n} C_\ell(\boldsymbol{\xi} + w_j \boldsymbol{\nu} p) \equiv C_\ell(\boldsymbol{\xi}) \pmod{p^2}. \tag{5.5}$$

Here we use $\boldsymbol{\xi} + w_j \boldsymbol{\nu} p$ to denote the vector $(\xi_1 + w_j \nu_1 p, \ldots, \xi_m + w_j \nu_m p)$.

Remark 5.1 We only require the mod p^2 supercongruence in this paper, but we note that the below proof can be adapted to prove the supercongruence holds modulo $p^{\min\{n,4\}}$ for $p \geq 11$. Additionally, we have observed computationally that the supercongruence often holds modulo p^n. Proving this using our approach would require a strengthening of Theorem 5.2.

Proof First, we show that we can reduce to the case that $\ell = m$. If $\ell < m$, then for each $\ell + 1 \leq i \leq m$ the reflection formula (5.2) for Γ_p gives us

$$\frac{1}{\Gamma_p(\xi_i)} = (-1)^{x_i} \Gamma_p(1 - \xi_i), \qquad \frac{1}{\Gamma_p(\xi_i + w_j \nu_i p)} = (-1)^{x_i} \Gamma_p(1 - \xi_i - w_j \nu_i p),$$

where x_i is the unique integer in $\{1, \ldots, p\}$ such that $\xi_i \equiv x_i \pmod{p}$. Define

$$\hat{\boldsymbol{\xi}} = \{\xi_1, \xi_2, \ldots, \xi_\ell, 1 - \xi_{\ell+1}, \ldots, 1 - \xi_m\}$$

$$\boldsymbol{\xi} + \widehat{w_j}\boldsymbol{\nu}p = \left\{\xi_1 + w_j\nu_i p, \ldots, \xi_\ell + w_j\nu_\ell p, 1 - \xi_{\ell+1} - w_j\nu_{\ell+1}p, \ldots, 1 - \xi_m - w_j\nu_m p\right\}.$$

Assuming (5.5) holds for $\ell = m$, then modulo p^2

$$\frac{1}{n}\sum_{i=1}^{n} C_\ell(\boldsymbol{\xi}+w_j\boldsymbol{\nu}p) = \frac{(-1)^{\sum_{i=\ell+1}^{m} x_i}}{n}\sum_{i=1}^{n} C_m(\boldsymbol{\xi} + \widehat{w_j}\boldsymbol{\nu}p) \equiv (-1)^{\sum_{i=\ell+1}^{m} x_i} C_m(\hat{\boldsymbol{\xi}}) = C_\ell(\boldsymbol{\xi}).$$

Thus, we need only prove the case $\ell = m$. As such, we suppress the subscript and simply write C to denote C_m for the remainder of the proof. For each $1 \le j \le n$ the definition of C yields

$$C(\boldsymbol{\xi} + w_j\boldsymbol{\nu}p) = \prod_{i=1}^{m} \Gamma_p(\xi_i + w_j\nu_i p).$$

Using Theorem 5.2, each term of the product can be approximated p-adically by

$$\Gamma_p(\xi_i + w_j\nu_i p) \equiv \Gamma_p(\xi_i)(1 + G_1(\xi_i)w_j\nu_i p) \pmod{p^2}.$$

Thus,

$$\begin{aligned} C(\boldsymbol{\xi} + w_j\boldsymbol{\nu}p) &\equiv \prod_{i=1}^{n} \Gamma_p(\xi_i)(1 + G_1(\xi_i)w_j\nu_i p) \pmod{p^2} \\ &\equiv C(\boldsymbol{\xi})\left(1 + \sum_{i=1}^{n} G_1(\xi_i)w_j\nu_i p\right) \pmod{p^2}. \end{aligned}$$

Hence,

$$\frac{1}{n}\sum_{j=1}^{n} C(\boldsymbol{\xi} + w_j\boldsymbol{\nu}p) \equiv C(\boldsymbol{\xi})\left(1 + \frac{1}{n}\sum_{1\le i,j\le n} G_1(\xi_i)w_j\nu_i p\right) \pmod{p^2}.$$

Thus, our desired congruence holds if

$$\frac{1}{n}\sum_{1\le i,j\le n} G_1(\xi_i)w_j\nu_i \equiv 0 \pmod{p}.$$

This follows from the assumption that the sum of the w_j's is zero. □

Remark 5.2 As a particular example, suppose that $n \mid p-1$ so that there exists a primitive n^{th} root of unity $\zeta_n \in \mathbb{Z}_p$. Then we can take $w = (1, \zeta_n, \zeta_n^2, \ldots, \zeta_n^{n-1})$.

5.2 A well-poised primitive ${}_4F_3(-1)$ supercongruence

We now turn our attention to the first supercongruence (1.3) in Theorem 1.3, which we restate below as Proposition 5.4 for the reader's convenience. Recall that, for $1 < j < 11$ we define $M = \operatorname{lcm}(4, D)$ for $D = 24/\gcd(j, 24)$, $r = \frac{j}{12}$, and

$$HD_4(r) = \left\{\left\{r, r, \frac{1}{2}, \frac{1}{2}\right\}, \left\{1, 1, r + \frac{1}{2}, r + \frac{1}{2}\right\}\right\}.$$

As $HD_4(r)$ is not defined over $\mathbb{Q}$ when $r \neq \frac{1}{2}$, the p-adic behavior is different for $1 \le j \le 6$ than it is for $6 < j < 12$. In this section, we only consider the case $j \le 6$. We expect similar results will hold for $7 \le j \le 11$.

Proposition 5.4 *Let* $1 \le j \le 6$ *and set* $r = j/12$. *For* $p \equiv 1 \pmod{M}$, *we have*

$$\Omega_{r,\mathbb{Q}_p} F\left(HD_4\left(r\right); -1\right)_{p-1} \equiv \left(a_p(f_{2,D}^{\sharp})\right)_0 a_p(f_{3,D}^{\sharp}) \pmod{p^2},$$

where $\Omega_{r,\mathbb{Q}_p} = \frac{\Gamma_p(r)^2}{\Gamma_p(r+1/2)^2}$, $f_{3,D}^{\sharp}$ *is given in Table 1 and* $\left(a_p(f_{2,D}^{\sharp})\right)_0$ *is given in Equation* (4.9).

We will obtain this by applying Proposition 5.3 to classical hypergeometric identities. In particular, we will use the ${}_4F_3$ case of Whipple's formula (4.1) as well as the following classical Kummer relation (see Corollary 3.3.5 of [3]) which holds whenever both sides converge.

$${}_3F_2\left[\begin{matrix} a & b & c \\ & d & e \end{matrix}; 1\right] = \frac{\Gamma(e)\Gamma(d+e-a-b-c)}{\Gamma(e-a)\Gamma(d+e-b-c)}\, {}_3F_2\left[\begin{matrix} a & d-b & d-c \\ & d & d+e-b-c \end{matrix}; 1\right]. \tag{5.6}$$

We first prove the following lemma using Proposition 5.3 which will be a necessary step in reducing $F(HD_4(r); -1)$ to a form where Theorem 2.3 of [1] can be applied.

Lemma 5.5 *Let* $1 \le j \le 6$ *and* $r = j/12$. *If* $p \equiv 1 \pmod{M}$, *then modulo* p^2,

$$p \cdot {}_3F_2\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & 1-\frac{r}{2} \\ & 1 & 1+\frac{r}{2} \end{matrix}; 1\right]_{p-1} \equiv \frac{-\Gamma_p\left(\frac{r}{2}\right)\Gamma_p(r)}{\Gamma_p\left(\frac{1+r}{2}\right)\Gamma_p\left(r+\frac{1}{2}\right)}\, {}_3F_2\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & \frac{r}{2} \\ & 1 & r+\frac{1}{2} \end{matrix}; 1\right]_{p-1}.$$

As hidden in the works [1,33,34], Theorem 1.2, and the proof of Proposition 5.4 below, we have different choices of length-3 data ($\mathbb{K}_2$-combinations) associated to the given Whipple formula used in this paper. Their $\mathbb{C}$-relations are recorded in the Appendix §6 with the classical identity Equation (5.6); their $\mathbb{F}_q$-relations are obtained earlier by Greene [15] and our explicit method [1]. The above Lemma can be thought of as a specialized p-adic version of Equation (5.6). We will leave the full p-adic generalization of this formula to interested readers.

Proof With $x = \pm 1$, we take $a = (1-xp)/2$, $b = \left(1-\frac{r}{2}\right)(1+p)$, $c = 1/2 - xp$, $d = 1 - xp/2$, and $e = 1 + r/2$ in (5.6). This gives

$$\begin{aligned} &{}_3F_2\left[\begin{matrix} \frac{1-xp}{2} & \left(1-\frac{r}{2}\right)(1+xp) & \frac{1}{2}-xp \\ & 1-\frac{xp}{2} & 1+\frac{r}{2} \end{matrix}; 1\right] \\ &= \frac{\Gamma\left(1+\frac{r}{2}\right)\Gamma\left(r+\frac{xpr}{2}\right)}{\Gamma\left(\frac{1+r+xp}{2}\right)\Gamma\left(r+\frac{1+xp(r-1)}{2}\right)}\, {}_3F_2\left[\begin{matrix} \frac{1-xp}{2} & \frac{1+xp}{2} & \frac{r+(r-3)xp}{2} \\ & 1-\frac{xp}{2} & r+\frac{1+xp(r-1)}{2} \end{matrix}; 1\right]. \end{aligned} \tag{5.7}$$

First, we observe that the ${}_3F_2$ on the right-hand side terminates at $(p-1)/2$ for both choices of x, as we will have the negative integer $(1-p)/2$ appearing in the upper parameters in each case. On the left-hand side, our hypergeometric series similarly terminates at $(p-1)/2$ for $x = 1$, but does not terminate until $(p-1)(1-r/2)$ for $x = -1$. Note that this index is an integer as the condition $p \equiv 1 \pmod{M}$ guarantees that the denominator of $r/2$ divides $p-1$. One can quickly see that the coefficients of the series will be divisible by p^2 for $(p-1)/2 < k \le (p-1)(1-r/2)$ and so the corresponding portion of the sum will

vanish modulo p^2. As such, we assume that our index k lies in the range $\{0, \dots, (p-1)/2\}$ throughout.

Beginning with the left-hand side of (5.7), we use (5.3) to rewrite the coefficients of the series as

$$\frac{\left(\frac{1-xp}{2}\right)_k \left(\left(1-\frac{r}{2}\right)(1+xp)\right)_k \left(\frac{1}{2}-xp\right)_k}{(1)_k \left(1-\frac{xp}{2}\right)_k \left(1+\frac{r}{2}\right)_k} = -\left(\frac{rp}{2}\right)^{-\nu\left(k,\frac{r(p-1)}{2}-1\right)} \times \Gamma_p\left(\frac{\frac{1-xp}{2}+k, \left(1-\frac{r}{2}\right)(1+xp)+k, \frac{1}{2}-xp+k, 1-\frac{xp}{2}, 1+\frac{r}{2}}{\frac{1-xp}{2}, \left(1-\frac{r}{2}\right)(1+xp), \frac{1}{2}-xp, k+1, k+1-\frac{xp}{2}, k+1+\frac{r}{2}}\right).$$

In the notation of Proposition 5.3, we have

$$p \cdot {}_3F_2\left[\begin{matrix}\frac{1-xp}{2} & \left(1-\frac{r}{2}\right)(1+xp) & \frac{1}{2}-xp \\ & 1-\frac{xp}{2} & 1+\frac{r}{2}\end{matrix}; 1\right] \equiv -p\sum_{k=0}^{\frac{p-1}{2}} C_5(\boldsymbol{\xi}^{(1)}(k)+\boldsymbol{\nu}^{(1)}xp)\left(\frac{rp}{2}\right)^{-\nu\left(k,\frac{r(p-1)}{2}-1\right)} \pmod{p^2}, \tag{5.8}$$

where

$$\boldsymbol{\xi}^{(1)}(k) := \left(\frac{1}{2}+k, 1-\frac{r}{2}+k, \frac{1}{2}+k, 1, 1+\frac{r}{2}, \frac{1}{2}, 1-\frac{r}{2}, \frac{1}{2}, k+1, k+1, k+1+\frac{r}{2}\right)$$

and

$$\boldsymbol{\nu}^{(1)} := \left(\frac{-1}{2}, 1+\frac{r}{2}, -1, \frac{-1}{2}, 0, \frac{-1}{2}, 1-\frac{r}{2}, -1, 0, \frac{-1}{2}, 0\right).$$

On the right-hand side, we begin with the quotient of Γ-values. By (5.4), we have

$$\frac{\Gamma\left(1+\frac{r}{2}\right)\Gamma\left(r+\frac{xpr}{2}\right)}{\Gamma\left(\frac{1+r+xp}{2}\right)\Gamma\left(r+\frac{1+xp(r-1)}{2}\right)} = \frac{\Gamma_p\left(1+\frac{r}{2}\right)\Gamma_p\left(r+\frac{xpr}{2}\right)}{\Gamma_p\left(\frac{1+r+xp}{2}\right)\Gamma_p\left(r+\frac{1+xp(r-1)}{2}\right)} \cdot \frac{2}{pr}.$$

The coefficients of the ${}_3F_2$ can be expressed in terms of Γ_p as before, which yields

$$p\frac{\Gamma\left(1+\frac{r}{2}\right)\Gamma\left(r+\frac{xpr}{2}\right)}{\Gamma\left(\frac{1+r+xp}{2}\right)\Gamma\left(r+\frac{1+xp(r-1)}{2}\right)} {}_3F_2\left[\begin{matrix}\frac{1-xp}{2} & \frac{1+xp}{2} & \frac{r+(r-3)xp}{2} \\ & 1-\frac{xp}{2} & r+\frac{1+xp(r-1)}{2}\end{matrix}; 1\right] = -p\sum_{k=0}^{\frac{p-1}{2}} C_6\left(\boldsymbol{\xi}^{(2)}(k)+\boldsymbol{\nu}^{(2)}xp\right)\cdot\frac{2}{pr}\cdot\left(\frac{p[(r-3)x+r]}{2}\right)^{\nu\left(k,\frac{(p-1)r}{2}\right)} \tag{5.9}$$

where

$$\boldsymbol{\xi}^{(2)}(k) = \left(1+\frac{r}{2}, r, \frac{1}{2}+k, \frac{1}{2}+k, \frac{r}{2}+k, 1, \frac{1+r}{2}, \frac{1}{2}, \frac{1}{2}, \frac{r}{2}, k+1, k+1, r+k+\frac{1}{2}\right)$$

and

$$\boldsymbol{\nu}^{(2)} = \left(0, \frac{r}{2}, \frac{-1}{2}, \frac{r-3}{2}, \frac{1}{2}, \frac{-1}{2}, \frac{1}{2}, \frac{-1}{2}, \frac{1}{2}, \frac{r-3}{2}, 0, \frac{-1}{2}, \frac{r-1}{2}\right).$$

In this case we have an equality rather than only a congruence modulo p^2 due to the fact that the ${}_3F_2$ on the right-hand side always terminates at $(p-1)/2$. Combining (5.8) and (5.9), we may reinterpret (5.7) p-adically as

$$\begin{aligned} &-p\sum_{k=0}^{\frac{p-1}{2}} C_5(\boldsymbol{\xi}^{(1)}(k) + \boldsymbol{\nu}^{(1)}xp)\left(\frac{rp}{2}\right)^{-\nu\left(k,\frac{r(p-1)}{2}-1\right)} \\ &\quad\equiv -p\sum_{k=0}^{\frac{p-1}{2}} C_6\left(\boldsymbol{\xi}^{(2)}(k) + \boldsymbol{\nu}^{(2)}xp\right)\cdot\frac{2}{pr}\cdot\left(\frac{p[(r-3)x+r]}{2}\right)^{\nu\left(k,\frac{(p-1)r}{2}\right)} \pmod{p^2}. \end{aligned} \tag{5.10}$$

For ease of notation, we let LHS(x) and RHS(x) denote the left and right-hand sides of (5.10), respectively, viewed as functions of x. As the congruence holds for $x = \pm 1$, we can average both sides over the two values of x as follows. We have

$$\frac{\mathrm{LHS}(1) + \mathrm{LHS}(-1)}{2} \equiv \frac{\mathrm{RHS}(1) + \mathrm{RHS}(-1)}{2} \pmod{p^2}. \tag{5.11}$$

On the left-hand side, we use Proposition 5.3 with $n = 2$ to conclude

$$\begin{aligned} \frac{\mathrm{LHS}(1) + \mathrm{LHS}(-1)}{2} = -p\sum_{k=0}^{\frac{p-1}{2}} &\left(\frac{C_5(\boldsymbol{\xi}^{(1)}(k) + \boldsymbol{\nu}^{(1)}xp) + C_5(\xi^{(1)}(k) - \boldsymbol{\nu}^{(1)}xp)}{2}\right) \\ &\times\left(\frac{rp}{2}\right)^{-\nu\left(k,\frac{r(p-1)}{2}-1\right)} \end{aligned}$$

Thus, modulo p^2,

$$\begin{aligned} \frac{\mathrm{LHS}(1) + \mathrm{LHS}(-1)}{2} &\equiv -p\sum_{k=0}^{\frac{p-1}{2}} C_5(\boldsymbol{\xi}^{(1)}(k))\left(\frac{rp}{2}\right)^{-\nu\left(k,\frac{r(p-1)}{2}-1\right)} \\ &\equiv p\sum_{k=0}^{\frac{p-1}{2}} \Gamma_p\left(\frac{\frac{1}{2}+k, 1-\frac{r}{2}+k, \frac{1}{2}+k, 1+\frac{r}{2}}{\frac{1}{2}, 1-\frac{r}{2}, \frac{1}{2}, k+1, k+1, k+1+\frac{r}{2}}\right)\left(\frac{2}{pr}\right)^{\nu\left(k,\frac{r(p-1)}{2}-1\right)} \\ &\equiv p\sum_{k=0}^{\frac{p-1}{2}} \frac{\left(\frac{1}{2}\right)_k^2\left(1-\frac{r}{2}\right)_k}{(1)_k^2\left(1+\frac{r}{2}\right)_k} \\ &\equiv p\,{}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & 1-\frac{r}{2} \\ 1 & 1+\frac{r}{2} & \end{matrix}\,;\,1\right]_{p-1}. \end{aligned} \tag{5.12}$$

The final congruence holds as the coefficients of this hypergeometric function are divisible by p for $(p-1)/2 < k \leq p-1$.

The right-hand side is more complicated, as we have

$$\begin{aligned}&\frac{\mathrm{RHS}(1)+\mathrm{RHS}(-1)}{2}\\&=\frac{-2}{r}\Bigg(\sum_{k=0}^{\frac{(p-1)r}{2}}\frac{C_6(\boldsymbol{\xi}^{(2)}(k)+\boldsymbol{\nu}^{(2)}p)+C_6(\boldsymbol{\xi}^{(2)}(k)-\boldsymbol{\nu}^{(2)}p)}{2}\\&\quad+\sum_{k=\frac{(p-1)r}{2}+1}^{\frac{p-1}{2}}\frac{\frac{p(2r-3)}{2}C_6(\boldsymbol{\xi}^{(2)}(k)+\boldsymbol{\nu}^{(2)}p)+\frac{3p}{2}C_6(\boldsymbol{\xi}^{(2)}(k)-\boldsymbol{\nu}^{(2)}p)}{2}\Bigg)\end{aligned}$$

For the first sum, we can simplify the inside to $C_6(\boldsymbol{\xi}^{(2)}(k))$ modulo p^2 as before. For the second sum, we note that we only require a congruence modulo p due to the additional multiple of p appearing. The congruences

$$C_6(\boldsymbol{\xi}^{(2)}(k)\pm\boldsymbol{\nu}^{(2)}p)\equiv C_6(\boldsymbol{\xi}^{(2)}(k))\pmod{p}$$

are immediate consequences of Theorem 5.2, and so

$$\begin{aligned}&\sum_{k=\frac{(p-1)r}{2}+1}^{\frac{p-1}{2}}\frac{\frac{p(2r-3)}{2}C_6(\boldsymbol{\xi}^{(2)}(k)+\boldsymbol{\nu}^{(2)}p)+\frac{3p}{2}C_6(\boldsymbol{\xi}^{(2)}(k)-\boldsymbol{\nu}^{(2)}p)}{2}\\&\equiv\sum_{k=\frac{(p-1)r}{2}+1}^{\frac{p-1}{2}}\frac{pr}{2}C_6(\boldsymbol{\xi}^{(2)}(k))\pmod{p^2}.\end{aligned}$$

Therefore, modulo p^2 we have

$$\begin{aligned}\frac{\mathrm{RHS}(1)+\mathrm{RHS}(-1)}{2}&\equiv\frac{-2}{r}\sum_{k=0}^{\frac{p-1}{2}}C_6(\boldsymbol{\xi}^{(2)}(k))(pr/2)^{\nu\left(k,\frac{(p-1)r}{2}\right)}\\&\equiv\frac{2}{r}\sum_{k=0}^{\frac{p-1}{2}}\Gamma_p\left(\frac{1+\frac{r}{2},r,\frac{1}{2}+k,\frac{1}{2}+k,\frac{r}{2}+k}{\frac{1+r}{2},\frac{1}{2},\frac{1}{2},\frac{r}{2},k+1,k+1,r+k+\frac{1}{2}}\right)\left(\frac{pr}{2}\right)^{\nu\left(k,\frac{(p-1)r}{2}\right)}\\&\equiv\frac{2}{r}\frac{\Gamma_p\left(1+\frac{r}{2}\right)\Gamma_p(r)}{\Gamma_p\left(\frac{1+r}{2}\right)\Gamma_p\left(r+\frac{1}{2}\right)}\sum_{k=0}^{\frac{(p-1)}{2}}\frac{\left(\frac{1}{2}\right)_k^2\left(\frac{r}{2}\right)_k}{k!^2\left(r+\frac{1}{2}\right)_k}\\&\equiv\frac{-\Gamma_p\left(\frac{r}{2}\right)\Gamma_p(r)}{\Gamma_p\left(\frac{1+r}{2}\right)\Gamma_p\left(r+\frac{1}{2}\right)}{}_3F_2\left[\begin{matrix}\frac{1}{2}&\frac{1}{2}&\frac{r}{2}\\1&r+\frac{1}{2}\end{matrix};1\right]_{p-1}.\end{aligned}\tag{5.13}$$

As with the left-hand side, we use the fact that the coefficients of this ${}_3F_2$ are divisible by p^2 for $(p-1)/2<k\le p-1$ to go from the truncation at $(p-1)/2$ to the truncation at $p-1$ in the congruence. Combining (5.11), (5.12), and (5.13) yields the claim. □

Proof of Proposition 5.4 We begin by taking $r_1=r_2=r$, $r_3=\frac{1+p}{2}$, and $r_4=\frac{1-p}{2}$ in (4.1) which gives

$${}_4F_3\left[\begin{matrix}r&r&\frac{1+p}{2}&\frac{1-p}{2}\\1&r+\frac{1-p}{2}&r+\frac{1+p}{2}\end{matrix};-1\right]=\frac{\Gamma\left(r+\frac{1-p}{2}\right)\Gamma\left(r+\frac{1+p}{2}\right)}{\Gamma(1+r)\Gamma(r)}\cdot{}_3F_2\left[\begin{matrix}\frac{1+p}{2}&\frac{1-p}{2}&1-\frac{r}{2}\\1&1+\frac{r}{2}\end{matrix};1\right].\tag{5.14}$$

Note that both sides terminate at $\frac{p-1}{2}$. By (5.4) and (5.1),

$$\frac{\Gamma\left(r+\frac{1-p}{2}\right)\Gamma\left(r+\frac{1+p}{2}\right)}{\Gamma(1+r)\Gamma(r)} = \frac{-\Gamma_p\left(r+\frac{1-p}{2}\right)\Gamma_p\left(r+\frac{1+p}{2}\right)}{\Gamma_p(r)^2}p.$$

We observe that both the Γ_p quotient above as well as the coefficients of each hypergeometric function appearing in (5.14) are symmetric in p and $-p$. Thus, we may use Proposition 5.3 to remove the p-linear terms modulo p^2, giving

$$F(HD_4(r);-1)_{p-1} \equiv \frac{-\Gamma_p\left(r+\frac{1}{2}\right)^2}{\Gamma_p(r)^2}pr\cdot {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & 1-\frac{r}{2}\\ & 1 & 1+\frac{r}{2}\end{matrix};\ 1\right]_{p-1} \pmod{p^2}.$$

To apply the supercongruence Theorem 2.3 of [1], we require the bottom right entry to lie between 1/2 and 1. This is not the case for us, but we can use the Kummer relation (5.6) to transform to a ${}_3F_2$ for which the supercongruence result of [1] does apply. Namely, by Lemma 5.5 we have

$$\begin{aligned} F(HD_4(r);-1)_{p-1} &\equiv \frac{-\Gamma_p\left(r+\frac{1}{2}\right)^2}{\Gamma_p(r)^2}\cdot p\ {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & 1-\frac{r}{2}\\ & 1 & 1+\frac{r}{2}\end{matrix};\ 1\right]_{p-1}\\ &\equiv \frac{\Gamma_p\left(r+\frac{1}{2}\right)\Gamma_p\left(\frac{r}{2}\right)}{\Gamma_p(r)\Gamma_p\left(\frac{1+r}{2}\right)}\cdot {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{r}{2}\\ & 1 & r+\frac{1}{2}\end{matrix};\ 1\right]_{p-1} \pmod{p^2}. \end{aligned}$$

Written this way, Theorem 2.3 of [1] then gives

$$F(HD_4(r);-1)_{p-1} \equiv \frac{\Gamma_p\left(r+\frac{1}{2}\right)\Gamma_p\left(\frac{r}{2}\right)}{\Gamma_p(r)\Gamma_p\left(\frac{1+r}{2}\right)}\cdot H_p\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{r}{2}\\ & 1 & r+\frac{1}{2}\end{matrix};\ 1;\overline{\omega}_p\right] \pmod{p^2}.$$

To ease notation, we will drop $\overline{\omega}_p$ from the character sum H_p below. Using the notation $HD_3(u,v)$ from (4.11) and continuing, we deduce

$$\begin{aligned} F(HD_4(r);-1)_{p-1} &\equiv \frac{\Gamma_p\left(r+\frac{1}{2}\right)\Gamma_p\left(\frac{r}{2}\right)}{\Gamma_p(r)\Gamma_p\left(\frac{1+r}{2}\right)}\cdot H_p\left(HD_3\left(\frac{r}{2},r+\frac{1}{2}\right);1\right) \pmod{p^2}\\ &\overset{(2.4)}{\equiv} \frac{\Gamma_p\left(r+\frac{1}{2}\right)^2}{\Gamma_p\left(\frac{1+r}{2}\right)^2\Gamma_p(r)}\mathbb{P}\left(HD_3\left(\frac{r}{2},r+\frac{1}{2}\right);1;\mathfrak{p}\right) \pmod{p^2}\\ &\equiv (-1)^{\frac{(p-1)r}{2}}\omega_p(4)^{-(p-1)r}\frac{\Gamma_p\left(r+\frac{1}{2}\right)^2}{\Gamma_p\left(\frac{1+r}{2}\right)^2\Gamma_p(r)}a_p(f_{3,D}^\sharp) \pmod{p^2}, \end{aligned}$$

where the final step is a result of (3.4) and [33, Section 3, Table 1]. (It can be derived by [15, (4.25)] and (4.12).)

Recalling $\Omega_{r,\mathbb{Q}_p} = \Gamma_p(r)^2/\Gamma_p(r+1/2)^2$ we conclude that

$$\begin{aligned} \Omega_{r,\mathbb{Q}_p}F(HD_4(r);-1)_{p-1} &\equiv (-1)^{\frac{(p-1)r}{2}}\omega_p(4)^{(1-p)r}\frac{\Gamma_p(r)^2}{\Gamma_p\left(r+\frac{1}{2}\right)^2}\frac{\Gamma_p\left(r+\frac{1}{2}\right)^2}{\Gamma_p\left(\frac{1+r}{2}\right)^2\Gamma_p(r)}a_p(f_{3,D}^\sharp)\\ &\equiv (-1)^{\frac{(p-1)r}{2}}\omega_p(1/4)^{(p-1)r}\frac{\Gamma_p(r)}{\Gamma_p\left(\frac{1+r}{2}\right)^2}a_p(f_{3,D}^\sharp) \end{aligned}$$

$$\overset{(4.9)}{\equiv} \left(a_p(f^{\sharp}_{2,D})\right)_0 a_p(f^{\sharp}_{3,D}) \pmod{p^2}.$$

□

5.3 A very well-poised ${}_5F_4(-1)$ supercongruence

We now turn our attention to the second congruence (1.4) of Theorem 1.3 which we restate in the following proposition:

Proposition 5.6 *For any prime $p \equiv 1 \pmod{M}$, $r = j/12$*

$$\Omega_{r,\mathbb{Q}_p} F(HD_5(r);-1)_{p-1} \equiv \left(a_p(f^{\sharp}_{2,D})\right)_1 a_p(f^{\sharp}_{3,D}) \pmod{p^2}$$

where $\left(a_p\left(f^{\sharp}_{2,D}\right)\right)_1 = p/\left(a_p\left(f^{\sharp}_{2,D}\right)\right)_0$.

Proof We begin by substituting

$$r_1 = r_2 = r, \quad r_3 = \frac{1+p}{2}, \quad r_4 = \frac{1-p}{2}$$

into (4.1). This gives us

$${}_5F_4\left[\begin{matrix} r & 1+\frac{r}{2} & r & \frac{1+p}{2} & \frac{1-p}{2} \\ & \frac{r}{2} & 1 & r+\frac{1-p}{2} & r+\frac{1+p}{2} \end{matrix}; -1\right] = \frac{\Gamma\left(r+\frac{1-p}{2}\right)\left(r+\frac{1+p}{2}\right)}{\Gamma(1+r)\,\Gamma(r)}\,{}_3F_2\left[\begin{matrix} \frac{1-r}{2} & \frac{1+p}{2} & \frac{1-p}{2} \\ & \frac{1+r}{2} & 1 \end{matrix}; 1\right].$$

Under this perturbation, both hypergeometric series now terminate at $(p-1)/2$. As in the previous proposition, the Γ-quotient can be expressed p-adically as

$$\frac{\Gamma\left(r+\frac{1-p}{2}\right)\Gamma\left(r+\frac{1+p}{2}\right)}{\Gamma(1+r)\Gamma(r)} = \frac{-\Gamma_p\left(r+\frac{1-p}{2}\right)\Gamma_p\left(r+\frac{1+p}{2}\right)}{\Gamma_p(r)^2}p.$$

As in the previous proof, we may use Proposition 5.3 to remove the p-linear terms on both sides modulo p^2 before using Theorem 2.3 of [1] and (3.4) to relate to the corresponding hypergeometric character sum. Doing so gives us

$$\begin{aligned}
\Omega_{j,\mathbb{Q}_p} F(HD_5(r);-1)_{p-1} &= -p\ {}_3F_2\left[\begin{matrix} \frac{1-r}{2} & \frac{1}{2} & \frac{1}{2} \\ & \frac{1+r}{2} & 1 \end{matrix}; 1\right] \pmod{p^2} \\
&\equiv -pH_p\left[\begin{matrix} \frac{1}{2} & \frac{1}{2} & \frac{1-r}{2} \\ & 1 & \frac{1+r}{2} \end{matrix}; 1\right] \pmod{p^2} \\
&\equiv \frac{-\Gamma_p\left(\frac{1+r}{2}\right)}{\Gamma_p(r)\Gamma_p\left(\frac{1-r}{2}\right)}p\,\mathbb{P}\left(HD_3\left(\frac{1-r}{2},\frac{1+r}{2}\right);1;p\right) \pmod{p^2} \\
&\equiv -(-1)^{\frac{(p-1)r}{2}}\frac{\Gamma_p\left(\frac{1+r}{2}\right)^2}{\Gamma_p(r)}p\,\omega_p(4)^{(p-1)r}a_p\left(f^{\sharp}_{3,D}\right) \pmod{p^2} \\
&\equiv \left(a_p(f^{\sharp}_{2,D})\right)_1 a_p\left(f^{\sharp}_{3,D}\right) \pmod{p^2},
\end{aligned}$$

as was to be shown. □

Remark 5.3 1. In the $r = 1/2$ case, the approach of §2.5 of [23] can be used to show that this congruence in fact holds modulo p^3. The congruence does not hold beyond p^2 for the other cases.

2. In Part I of the series [1, (5.29)], we introduced the hypergeometric sum $E_{\mathrm{GK}}(HD;\lambda)$ as the potential obstruction to a mod p^2 supercongruence. With this, the supercongruence results in this paper can be rephrased as, for $p \equiv 1 \pmod{M}$,

$$E_{\mathrm{GK}}(HD_4(r);-1) \equiv \Omega_{r,\mathbb{Q}_p}^{-1} \cdot \left(a_p\left(f_{2,D}^{\sharp}\right)\right)_1 a_p\left(f_{3,D}^{\sharp}\right)/p \pmod{p}.$$

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6 Appendix

6.1 Some L-values of weight three CM modular forms in terms of ${}_3F_2(1)$

In the literature, hypergeometric values have been used to compute special L-values of modular forms (some discussion related to η-products includes [20,29,31,43, et al.]) and other arithmetic objects, such as Mahler measures of two-variable Laurent polynomials, as in [32] by Rogers–Zudilin. In this subsection, we record the special L-values of certain cuspforms related to the ${}_3F_2(HD_d^{(3)};1)$ values, where $HD_d^{(3)} := \{\{\frac{1}{2}, \frac{1}{d}, 1-\frac{1}{d}\}, \{1,1,1\}\}$ for $d = 2, 3, 4$, and 6. The CM cases of the G2 family not in this form will be treated in the next subsection.

Theorem 6.1

$${}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{2}\\ 1 & 1\end{matrix};\,1\right] = \frac{2}{\sqrt{3}}\,{}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{6} & \frac{5}{6}\\ 1 & 1\end{matrix};\,1\right] = \frac{8}{\pi}L(\eta(4\tau)^6,1) = \frac{16}{\pi^2}L(\eta(4\tau)^6,2);$$

$${}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{3} & \frac{2}{3}\\ 1 & 1\end{matrix};\,1\right] = \frac{6\sqrt{3}}{\pi}L(\eta(2\tau)^3\eta(6\tau)^3,1) = \frac{18}{\pi^2}L(\eta(2\tau)^3\eta(6\tau)^3,2).$$

Let $f_{8.3.d.a}(\tau) = \eta(\tau)^2\eta(2\tau)\eta(4\tau)\eta(8\tau)^2$.

$${}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{4} & \frac{3}{4}\\ 1 & 1\end{matrix};\,1\right] = \sqrt{2}\,{}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{2}\\ 1 & 1\end{matrix};\,-1\right] = \frac{12\sqrt{2}}{\pi}L(f_{8.3.d.a}(\tau),1) = \frac{24}{\pi^2}L(f_{8.3.d.a}(\tau),2).$$

Proof By a Bailey cubic hypergeometric transformation [3, pp. 185], we have

$${}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{6} & \frac{5}{6}\\ 1 & 1\end{matrix};\,1\right] = \frac{\sqrt{3}}{2}\,{}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{2}\\ 1 & 1\end{matrix};\,1\right].$$

The result follows as stated in §3.4. In the work of Osburn and Straub [29], they obtain the special value

$$ {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{2}\\ 1 & 1\end{matrix}; -1\right] = \frac{12}{\pi}L(f_{8.3.d.a}(\tau), 1).$$

By a quadratic transformation (or identities between the corresponding modular forms, see [1, Appendix] for example), one has

$$ {}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{4} & \frac{3}{4}\\ 1 & 1\end{matrix}; 1\right] = \sqrt{2}\,{}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & \frac{1}{2}\\ 1 & 1\end{matrix}; -1\right] = \frac{12\sqrt{2}}{\pi}L(f_{8.3.d.a}(\tau), 1) = \frac{24}{\pi^2}L(f_{8.3.d.a}(\tau), 2).$$

For the $d = 3$ case, see [16] for detail. □

6.2 From modular forms to new ${}_3P_2(1)$ identities

Assume (r, s) is in a Galois $\mathbb{K}_2$ orbit. As a byproduct of Lemma 3.5 and the fact that $\{\mathbb{K}_2(r_c, q_c)\}$ and $\{\mathbb{K}_2^*(r_c, q_c)\}$ are two bases for the corresponding invariant subspaces of the Hecke operators, one has identities among the ${}_3F_2(1)$-values, which may be induced from the classical hypergeometric identities. In the following, we provide an immediate corollary of (5.6) and less trivial types of identities. For a more detailed discussion and other types of relations, please see the work by Rosen [33].

For a shorthand notation, we let $P(r, s)$ denote the period value

$$ P(r, s) := \pi B(r, s-r)\,{}_3F_2\left[\begin{matrix}\frac{1}{2} & \frac{1}{2} & r\\ 1 & s\end{matrix}; 1\right]. \tag{6.1}$$

Using this notation, a specialized version of Equation (5.6) is as follows.

$$ P(r, s) = \sin(\pi r)\frac{B(s-r, r)^2}{B(s-1/2, 1/2)^2}P(1-r, s-r+1/2). \tag{6.2}$$

We first consider the cases when $m = 4$ and 8, which are CM cases.

Proposition 6.2 *When $m = 4$ and 8, we have*

$$\mathbb{K}_2(1/3, 2/3) = \mathbb{K}_2^*(1/3, 2/3) \text{ and } \mathbb{K}_2(2/3, 4/3) = \mathbb{K}_2^*(2/3, 4/3).$$

$$\begin{aligned}\mathbb{K}_2(1/6, 5/6)(12\tau) =& \mathbb{K}_2(1/3, 2/3)(6\tau) - 4\mathbb{K}_2(2/3, 4/3)(12\tau)\\ \mathbb{K}_2^*(1/6, 5/6)(6\tau) =& \mathbb{K}_2(2/3, 5/6)(6\tau) = 2\mathbb{K}_2(1/3, 2/3)(12\tau) - \mathbb{K}_2(2/3, 4/3)(6\tau)\\ \mathbb{K}_2^*(5/6, 7/6)(6\tau) =& \mathbb{K}_2(1/3, 7/6)(6\tau) = \mathbb{K}_2(1/3, 2/3)(6\tau) - 8\mathbb{K}_2(2/3, 4/3)(12\tau).\end{aligned}$$

The corresponding P-values are algebraically dependent as follows.

$$\begin{aligned}P(1/6, 5/6) =& 2^{-5/6}3^{-1/4}B(1/4, 1/4)^2 = 2^{1/6}3^{-1/4}B(1/4, 1/2)^2,\\ 2^{\frac{1}{3}}P(1/3, 2/3) =& (2+\sqrt{3})P(2/3, 4/3),\\ 2^{\frac{1}{3}}P(5/6, 7/6) =& P(1/6, 5/6) = (1+\sqrt{3})P(2/3, 4/3),\\ 2^{\frac{1}{3}}P(1/3, 7/6) =& (2-\sqrt{3})P(2/3, 5/6) = \sqrt{3}P(2/3, 4/3).\end{aligned}$$

Proof Comparing the q-expansions, we have

$$\mathbb{K}_2\left(1/6,5/6\right)(12\tau)=\mathbb{K}_2\left(1/3,2/3\right)(6\tau)-4\mathbb{K}_2\left(2/3,4/3\right)(12\tau)$$

and hence

$$\begin{aligned}L\left(\mathbb{K}_2\left(1/6,5/6\right)(12\tau),1\right)=&L\left(\mathbb{K}_2\left(1/3,2/3\right)(6\tau),1\right)-4L\left(\mathbb{K}_2\left(2/3,4/3\right)(12\tau),1\right)\\=&L\left(\mathbb{K}_2\left(1/3,2/3\right)(6\tau),1\right)-2L\left(\mathbb{K}_2\left(2/3,4/3\right)(6\tau),1\right).\end{aligned}$$

Other identities can be obtained in a similar way.

By Lemma 3.5, Corollary 6.2, $2^{\frac{1}{6}}\Gamma(1/3)\Gamma(5/6)=\sqrt{2\pi}\Gamma(2/3)$ which follows from the duplication formula of $\Gamma(\cdot)$, and $\pi=\Gamma(1/3)\Gamma(2/3)\sin(\pi/3)$ from the reflection formula of $\Gamma(\cdot)$, we derive the algebraic relations

$$2^{\frac{1}{3}}P\left(1/3,2/3\right)=(2+\sqrt{3})P\left(2/3,4/3\right)$$

and

$$P\left(1/6,5/6\right)=2^{1/3}P(1/3,2/3)-P(2/3,4/3)=(1+\sqrt{3})P\left(2/3,4/3\right).$$

To evaluate $P(1/6,5/6)$, we first use (5.6) and then apply Dixon's evaluation [41, (2.2)] and properties of the gamma function.

$${}_3F_2\left[\begin{matrix}\frac{1}{2}&\frac{1}{2}&\frac{1}{6}\\ \frac{5}{6}&1\end{matrix};\,1\right]=\frac{\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{7}{6}\right)}\,{}_3F_2\left[\begin{matrix}\frac{1}{2}&\frac{1}{3}&\frac{2}{3}\\ \frac{7}{6}&\frac{5}{6}\end{matrix};\,1\right]=\frac{2^{3/2}}{3^{1/4}\pi^2}\Gamma\left(\begin{matrix}\frac{2}{3},\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4}\\ \frac{1}{2},\frac{1}{6}\end{matrix}\right),$$

which is equivalent to our claim. □

The $P\left(1/3,2/3\right)$-value can also be obtained by [3, Theorem 3.5.5(i)] with $c=1/3$ followed by the reflection and multiplication formulae of the Γ-function.

For other cases, we have the following relations between $\mathbb{K}_2^*(m/24,1-m/24)=\mathbb{K}_2(1-m/12,1-m/24)$ and $\mathbb{K}_2(mj/24,1-mj/24)$ in the same Hecke space, which lead to the P-identities.

Proposition 6.3 *For a fixed $m=1,2,3$, let $j\in(\mathbb{Z}/(12/m)\mathbb{Z})^\times$.*

$$\begin{aligned}&\mathbb{K}_2\left(\frac{mj}{12},\frac{3}{4}\pm\frac{1}{4}+\frac{mj}{24}\right)\left(\frac{24}{m}\tau\right)\\&\quad=\mathbb{K}_2\left(\frac{mj}{24},1-\frac{mj}{24}\right)\left(\frac{48}{m}\tau\right)\mp4\mathbb{K}_2\left(\frac{1}{2}+\frac{mj}{24},\frac{3}{2}-\frac{mj}{24}\right)\left(\frac{48}{m}\tau\right),\end{aligned}$$

here we use $\mp$ to denote the opposite sign choice from $\pm$. In terms of P-values,

$$2^{1-mj/6}P\left(\frac{mj}{12},\frac{3}{4}\pm\frac{1}{4}+\frac{mj}{24}\right)=P\left(\frac{mj}{24},1-\frac{mj}{24}\right)\mp P\left(\frac{1}{2}+\frac{mj}{24},\frac{3}{2}-\frac{mj}{24}\right).$$

Proof Below we let $t(\tau)$ be the Hauptmodul on $\Gamma_0(8)$ such that

$$\lambda(2\tau) = 16t(\tau)(1-4t(\tau)).$$

The identities between the modular forms follow from our construction and the classical results

$$2^{4r-1}\mathbb{K}_2(r,s)(\tau) = \left(\frac{\lambda}{1-\lambda}\right)^r (1-\lambda)^s\theta_3^6(\tau),$$

$$\frac{\lambda(\tau)^2}{1-\lambda(\tau)} = \frac{16\lambda(2\tau)}{(1-\lambda(2\tau))^2} \quad \text{and} \quad (1-4t(\tau))(1-\lambda(2\tau))^{\frac{1}{2}}\theta_3(2\tau)^6 = (1-\lambda(4\tau))\theta_3(4\tau)^6.$$

For a given m, we fix the choices $\lambda^{m/24}$ and $(1-\lambda)^{m/24}$. Similarly, we take $(1-\lambda(2\tau))^{1/2} = 1-8t(\tau)$. Then

$$\begin{aligned}
&2\cdot 2^{\frac{mj}{12}}\mathbb{K}_2\left(mj/24, 1-mj/24\right)(48\tau/m)\\
&\quad= 4\cdot 2^{4\cdot\frac{mj}{24}}\left(\frac{\lambda}{(1-\lambda)^2}\right)^{mj/24}(1-\lambda)\theta_3^6\left(48\tau/m\right)\\
&\quad= 4\left(\frac{\lambda^2}{1-\lambda}\right)^{mj/24}(1-\lambda)^{\frac{1}{2}}\theta_3^6\left(24\tau/m\right)\left(1-4t(12\tau/m)\right)\\
&\quad= 2\left(\frac{\lambda^2}{1-\lambda}\right)^{mj/24}(1-\lambda)^{\frac{1}{2}}(1+(1-\lambda)^{\frac{1}{2}})\theta_3^6\left(24\tau/m\right)\\
&\quad= 2\left(\frac{\lambda^2}{(1-\lambda)^2}\right)^{mj/24}(1-\lambda)^{\frac{mj}{24}}(1-\lambda)^{\frac{1}{2}}(1+(1-\lambda)^{\frac{1}{2}})\theta_3^6\left(24\tau/m\right)\\
&\quad= 2^{\frac{mj}{12}}\mathbb{K}_2\left(\frac{mj}{12}, \frac{1}{2}+\frac{mj}{24}\right)(24\tau/m) + 2^{\frac{mj}{12}}\mathbb{K}_2\left(\frac{mj}{12}, 1+\frac{mj}{24}\right)(24\tau/m).
\end{aligned}$$

By the same argument, we have

$$8\mathbb{K}_2\left(\frac{1}{2}+\frac{mj}{24}, \frac{3}{2}-\frac{mj}{24}\right)\left(\frac{48}{m}\tau\right) = \mathbb{K}_2\left(\frac{mj}{12}, \frac{1}{2}+\frac{mj}{24}\right)\left(\frac{24}{m}\tau\right) - \mathbb{K}_2\left(\frac{mj}{12}, 1+\frac{mj}{24}\right)\left(\frac{24}{m}\tau\right).$$

These equations are equivalent to the identities stated in the proposition and the corresponding P-values follow from Lemma 3.5. We will omit the details. □

6.3 Special L-values of $f_{3,D}^{\sharp}$ and some of their twists

Using identities in the previous subsection, one obtains the following applications. For a given Hecke eigenform $f^{\sharp}$, the relationships listed below induce the functional equations relating $L(f^{\sharp}, 2)$ and $L(f^{\sharp}\otimes\chi, 1)$ for an odd inner twist χ modulo N, which is guaranteed by the result of Shimura [35, Theorem 1] that

$$L(f^{\sharp}, 2) \in \mathfrak{g}(\chi)\pi i L(f^{\sharp}\otimes\chi, 1)\cdot\mathbb{Q}(a_n(f^{\sharp}) : n \geq 1),$$

if $\chi(-1) = -1$, where $\mathfrak{g}(\chi) = \sum_{j=0}^{N-1} e^{2\pi ij/N}\chi(j)$. In the end of this section, we provide the L-values of the chosen newforms $f^{\sharp}$ and some functional equations relating $L(f^{\sharp}, 2)$ and $L(f^{\sharp}\otimes\chi, 1)$.

Corollary 6.4 *For $m = 8$, let $f^\sharp$ be the newform $f_{36.3.d.a}(\tau) = \mathbb{K}_2(1/3, 2/3)(6\tau) - 2\mathbb{K}_2(2/3, 4/3)(6\tau)$. Then*

$$(1+\sqrt{3})P(2/3, 4/3) = \frac{B(1/4, 1/4)^2}{2^{\frac{5}{6}}3^{\frac{1}{4}}} = 6\pi \cdot 2^{2/3} \cdot L(f^\sharp, 1) = 18 \cdot 2^{2/3} \cdot L(f^\sharp, 2).$$

This result follows from Lemma 3.5 and Proposition 6.2.

Corollary 6.5 *The transcendental part of the Petersson inner product of $f_{36.3.d.a}$ or $f_{36.3.d.b}$ is $B(1/4, 1/4)^4/\pi^5$.*

This result follows from the work of Shimura [35, Theorem 4], Proposition 6.2 and the fact $f_{36.3.d.b} = f_{36.3.d.a} \otimes \chi_{-3}$.

Corollary 6.6 *Let $m = 3$ and $f^\sharp$ be the newform given in Table 2. Then*

$$L(f^\sharp, 1) = \frac{1}{16\pi}\left((2+\sqrt{2})(1+i)P(1/4, 5/8) + \sqrt{2}(1-i)P(1/4, 9/8)\right),$$

$$\begin{aligned} L(f^\sharp, 2) &= \frac{1}{64}\left((2+\sqrt{2})P(1/4, 5/8) + \sqrt{2}iP(1/4, 9/8)\right) \\ &= \frac{1+i}{8}\pi \cdot L(f^\sharp \otimes \chi_{-2}, 1) = \frac{(i-1)(1+\sqrt{2})}{8}\pi \cdot L(f^\sharp \otimes \chi_{-1}, 1). \end{aligned}$$

Proof According to Corollary 6.2 and Proposition 6.3, we have

$$\begin{gathered} P(3/4, 7/8) = (2+\sqrt{2})P(1/4, 5/8), \quad P(3/4, 11/8) = (2-\sqrt{2})P(1/4, 9/8), \\ \sqrt{2}P(1/8, 7/8) = P(1/4, 5/8) + P(1/4, 9/8), \quad \sqrt{2}P(5/8, 11/8) = P(1/4, 5/8) - P(1/4, 9/8), \\ 2P(3/8, 5/8) = (\sqrt{2}+1)P(1/4, 5/8) + (\sqrt{2}-1)P(1/4, 9/8), \\ 2P(7/8, 9/8) = (\sqrt{2}+1)P(1/4, 5/8) - (\sqrt{2}-1)P(1/4, 9/8). \end{gathered}$$

Hence by the integral formula (Lemma 3.5) and above identities,

$$\begin{aligned} 16\pi L(f^\sharp, 1) &= \sqrt{2}\left(P(1/8, 7/8) + \sqrt{2}P(3/8, 5/8) + iP(5/8, 11/8) + \sqrt{2}iP(7/8, 9/8)\right) \\ &= (2+\sqrt{2})(1+i)P(1/4, 5/8) + \sqrt{2}(1-i)P(1/4, 9/8), \end{aligned}$$

and

$$\begin{aligned} 16^2 L(f^\sharp, 2) &= 2^{3/2}P(3/4, 7/8) + 4P(1/4, 5/8) + 2\sqrt{2}iP(3/4, 11/8) + 4iP(1/4, 9/8) \\ &= 4(2+\sqrt{2})P(1/4, 5/8) + 4\sqrt{2}iP(1/4, 9/8). \end{aligned}$$

On the other hand, one can simplify $L(f^\sharp \otimes \chi_{-1}, 1)$ and $L(f^\sharp \otimes \chi_{-2}, 1)$ as follows.

$$\begin{aligned} \frac{16\pi}{\sqrt{2}} L(f^\sharp \otimes \chi_{-1}, 1) &= P(1/8, 7/8) + iP(5/8, 11/8) - \sqrt{2}P(3/8, 5/8) - \sqrt{2}iP(7/8, 9/8) \\ &= -(1+i)P(1/4, 5/8) + (\sqrt{2}-1)(1-i)P(1/4, 9/8) \\ &= \frac{(1-\sqrt{2})(1+i)}{\sqrt{2}}\left((2+\sqrt{2})P(1/4, 5/8) + \sqrt{2}iP(1/4, 9/8\right) \\ &= \frac{(1-\sqrt{2})(1+i)}{\sqrt{2}} 64 \cdot L(f^\sharp, 2); \end{aligned}$$

$$\begin{aligned}\frac{16\pi}{\sqrt{2}}L(f^\sharp\otimes\chi_{-2},1) &= P(1/8,7/8)-iP(5/8,11/8)+\sqrt{2}P(3/8,5/8)-\sqrt{2}iP(7/8,9/8)\\ &= (\sqrt{2}+1)(1-i)P(1/4,5/8)+i(1-i)P(1/4,9/8)\\ &= \frac{(1-i)}{\sqrt{2}}64\cdot L(f^\sharp,2),\end{aligned}$$

which give the identities in the corollary. □

Other functional equations listed in the table can be deduced in a similar manner; for interested readers, please see [33] (See Tables 4 and 5).

Table 4 G2 L-values in terms of $P(m/24)=P\left(m/24,\frac{24-m}{24}+\lfloor\frac{m}{12}\rfloor\right)$ defined in (6.1)

m	$L(f^\sharp,1)$
1	$\pi L(f_{1152.3.b.i}(\tau),1)=$ $\frac{1}{192\cdot 2^{\frac{1}{6}}}\left[8P(1/24)-4\beta_5P(7/24)-2\beta_2P(13/24)-\beta_7P(19/24)\right]$ $-\frac{1}{192\cdot 2^{\frac{5}{6}}}\left[8\beta_1P(5/24)+4\beta_3P(11/24)+2\beta_4P(17/24)+\beta_6P(23/24)\right]$
2	$\pi L(f_{288.3.g.a}(\tau),1)=\frac{1}{12\cdot 2^{\frac{1}{3}}}\left[P(1/12)+i\cdot P(7/12)\right]-\frac{1}{6\cdot 2^{\frac{2}{3}}}\left[P(5/12)+i\cdot P(11/12)\right]$
3	$\pi L(f_{128.3.d.c}(\tau),1)=\frac{1}{8\sqrt{2}}\left[P(1/8)+i\cdot P(5/8)\right]+\frac{1}{8}\left[P(3/8)+i\cdot P(7/8)\right]$
6	$\pi L(f_{32.3.c.a}(\tau),1)=\frac{1}{8}\left(P(1/4)+i\cdot P(3/4)\right)$
4, 8	$\pi L(f_{36.3.d.a},1)=\frac{\pi}{\sqrt{3}}L(f_{36.3.d.b},1)=\frac{B(1/4,1/4)^2}{2^{\frac{5}{2}}3^{\frac{5}{4}}}$

Here $\beta_1=2\sqrt{-7}$, $\beta_2=4\sqrt{-1}$, $\beta_3=4\sqrt{2}$, $\beta_4=8\sqrt{7}$, $\beta_5=2\sqrt{-14}$, $\beta_6=-16\sqrt{-2}$, $\beta_7=8\sqrt{14}$, and $B(x,y)$ is the Beta function

Table 5 G2 L-values at 2

m	$L(f^\sharp,2)$
1	$L(f^\sharp,2)=-\frac{1+i}{12}\pi\cdot L(f^\sharp\otimes\chi_{-2},1)$
2	$L(f_{288.3.g.a},2)=-\frac{1+i}{12}\pi\cdot L(f_{288.3.g.a}\otimes\chi_{-1},1)=\frac{1}{\sqrt{3}}\frac{1-i}{12}\pi\cdot L(f_{288.3.g.c},1)$
3	$L(f^\sharp,2)=\frac{1+i}{8}\pi\cdot L(f^\sharp\otimes\chi_{-2},1)=\frac{(i-1)(1+\sqrt{2})}{8}\pi\cdot L(f^\sharp\otimes\chi_{-1},1)$
6	$L(f^\sharp,2)=\frac{1+i}{4}\pi\cdot L(f^\sharp\otimes\chi_{-1},1)$
4, 8	$L(f^\sharp,2)=\frac{\pi}{3}\cdot L(f^\sharp,1)$

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